TP-LINK®

User Guide

TL-WN821N
Wireless N USB Adapter



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1910010343

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FCC STATEMENT



This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/ TV technician for help.

This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions:

1) This device may not cause harmful interference.
2) This device must accept any interference received, including interference that may cause undesired operation.

Any changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

Note: The manufacturer is not responsible for any radio or tv interference caused by unauthorized modifications to this equipment. Such modifications could void the user's authority to operate the equipment.

FCC RF Radiation Exposure Statement:

This device has been tested for compliance with FCC RF Exposure (SAR) limits in the typical laptop computer configuration and this device can be used in desktop or laptop computers. This device cannot be used with handheld PDAs (personal digital assistants). This device and its antenna must not be co-located or operated in conjunction with any other antenna or transmitter. SAR measurements are based on a 5mm spacing from the body and that compliance is achieved at that distance.

CE Mark Warning

CE 1588 ①

This is a class B product. In a domestic environment, this product may cause radio interference, in which case the user may be required to take adequate measures.

National restrictions

This device is intended for home and office use in all EU countries (and other countries following the EU directive 1999/5/EC) without any limitation except for the countries mentioned below:

Country	Restriction	Reason/remark
Bulgaria	None	General authorization required for outdoor use and public service
France	Outdoor use limited to 10 mW e.i.r.p. within the band 2454-2483.5 MHz	Military Radiolocation use. Refarming of the 2.4 GHz band has been ongoing in recent years to allow current relaxed regulation. Full implementation planned 2012
Italy	None	If used outside of own premises, general authorization is required
Luxembourg	None	General authorization required for network and service supply(not for spectrum)
Norway	Implemented	This subsection does not apply for the geographical area within a radius of 20 km from the centre of Ny-Ålesund
Russian Federation	None	Only for indoor applications

Note: Please don't use the product outdoors in France.

DECLARATION OF CONFORMITY

For the following equipment:

Product Description: **Wireless N USB Adapter**

Model No.: **TL-WN821N**

Trademark: **TP-LINK**

We declare under our own responsibility that the above products satisfy all the technical regulations applicable to the product within the scope of Council Directives:

Directives 1999/5/EC

The above product is in conformity with the following standards or other normative documents:

ETSI EN 300 328 V1.7.1: 2006

ETSI EN 301 489-1 V1.8.1:2008 & ETSI EN 301 489-17 V1.3.2:2008

EN60950-1:2006

EN62311:2008

Recommendation 1999/519/EC

Person is responsible for marking this declaration:



Yang Hongliang

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Package Contents

The following contents should be found in your box:

- One TL-WN821N Wireless N USB Adapter
- Quick Installation Guide
- One USB cable
- One Resource CD for TL-WN821N, including:
  - TP-LINK Wireless N Client Utility (TWCU) and Drivers
  - User Guide
  - Other Helpful Information

☞ **Note:**

Make sure that the package contains the above items. If any of the listed items are damaged or missing, please contact with your distributor.

Conventions:

The ‘Adapter’ mentioned in this user guide stands for TL-WN821N Wireless N USB Adapter without any explanations.

Chapter 1. Introduction

Thank you for choosing **TL-WN821N Wireless N USB Adapter**.

1.1 Overview of the product

The adapter is an 802.11n client device; it is designed to provide a high-speed and unrivaled wireless performance for your notebook and PC. With a faster wireless connection, you can get a better Internet experience, such as downloading, gaming, video streaming and so on.

With the 802.11n technology, higher throughput improvements using MIMO (multiple input, multiple output antennas), the TL-WN821N's auto-sensing capability allows high packet transfer rate of up to 300Mbps for maximum throughput. It has good capability on anti-jamming, it can also interoperate with other wireless (802.11b/g) products. The adapter supports WEP, WPA and WPA2 encryption to prevent outside intrusion and protect your personal information from being exposed.

The Quick Setup Wizard guides you step-by-step through the installation process; the TP-LINK Wireless N Client Utility helps you create a wireless connection immediately.

With unmatched wireless performance, reception, and security protection, the TL-WN821N is the best choice for easily adding or upgrading wireless connectivity.

1.2 Features

- Complies with IEEE802.11b, IEEE802.11g, IEEE802.11n standards
- Supports WPA/WPA2 data security, IEEE802.1x authentication, TKIP/AES encryption, WEP encryption
- Supports high rate of up to 300Mbps for maximum throughput, supports automatically adjust to lower speeds due to distance or other operating limitations
- Provides USB interface
- Supports Ad-Hoc and Infrastructure modes
- Good capability on anti-jamming
- Supports roaming between access points when configured in Infrastructure mode
- Ease to configure and provides monitoring information
- Supports Windows 2000, XP, Windows Vista and Windows 7.
- Built-in two internal antennas which are listed in a format of 2x2 for two receivers and two transmitters.

1.3 LED Status

Status	Working Status
On	The driver has not been installed.
Flashing Slowly	The driver has been installed but there is no data being transmitted or received.
Flashing Quickly	There is data being transmitted or received.

☞ **Note:**

When the card failed to connect to a wireless network, the LED may be off. Please choose your network and try to connect again.

Chapter 2. Installation Guide

2.1 Hardware Installation

There are two ways to install the Adapter:

1. Plug the Adapter directly to the USB port on your computer.
2. Connect the Adapter and your computer through the USB cable attached in package.

☞ **Note:**

The promoted Found New Hardware Wizard screen will pop up when the adapter is installed correctly. Click **Cancel**.

2.2 Software Installation

2.2.1 For Windows XP

The Setup steps for Windows 2000 and XP are similar with each other. This user guide takes Windows XP for example.

1. Insert the Resource CD into your CD-ROM drive, and then Figure 2-1 will appear. To continue, select the **TL-WN821N**. There will be a menu including **Install Driver & Utility**, **Install QSS** and **User Guide**.



Figure 2-1

2. Click **Install Driver & Utility** to load the following screen.

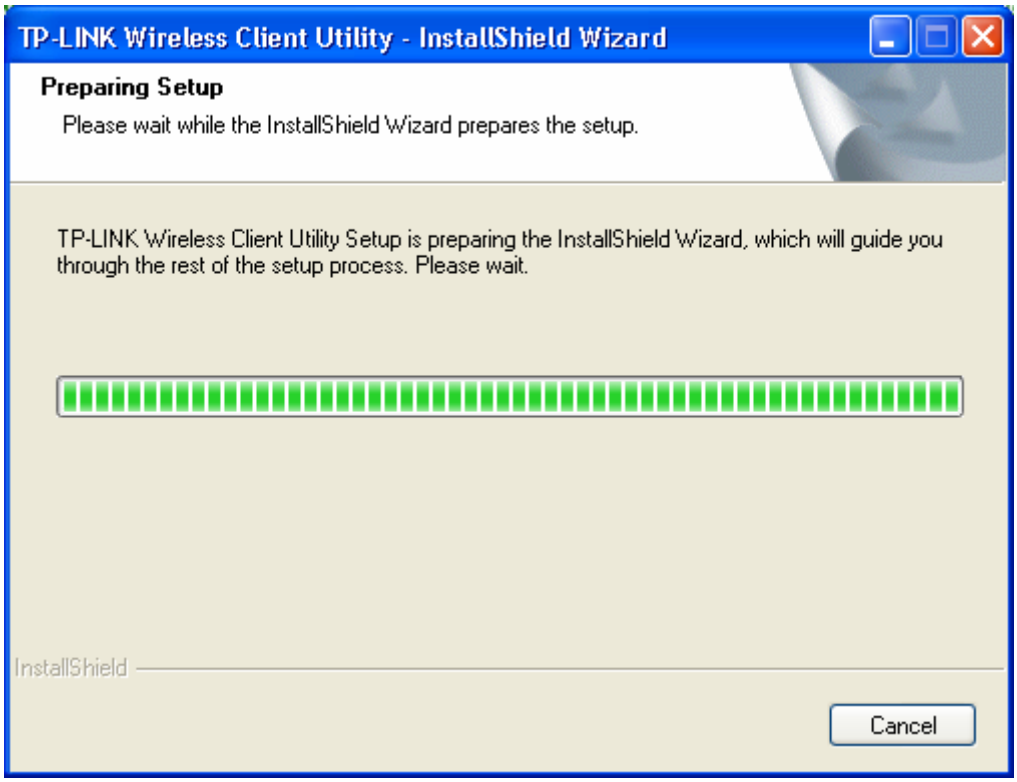


Figure 2-2

3. Soon, Figure 2-3 will display after a moment. Click **Next** to continue.

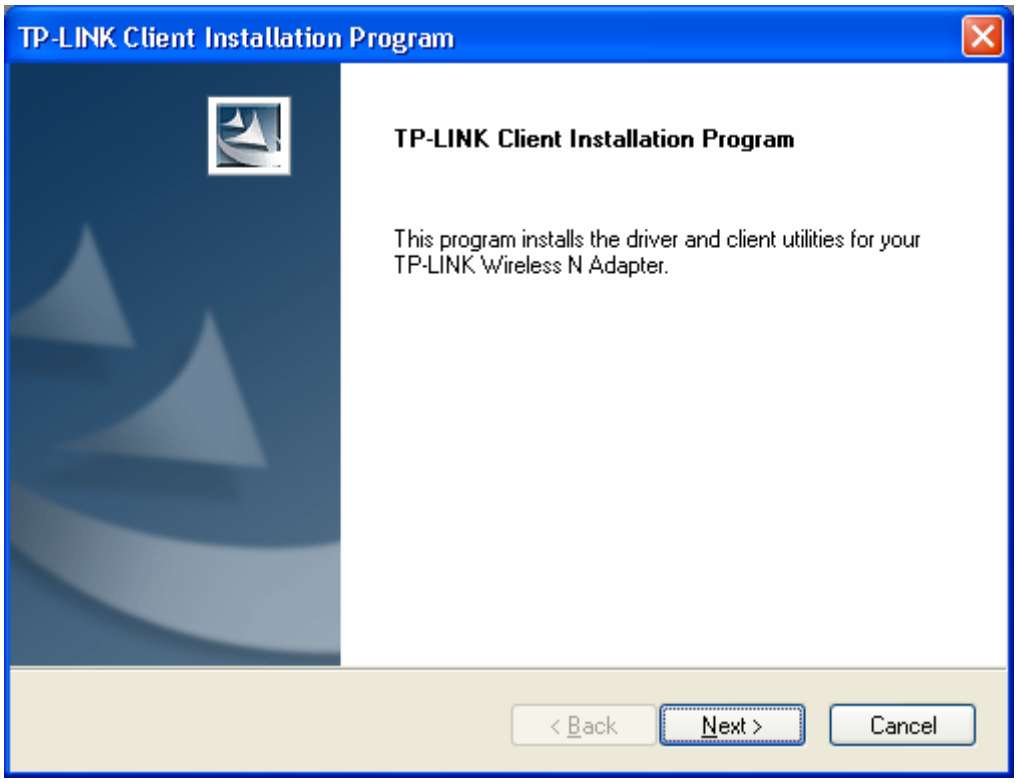


Figure 2-3

4. After that, you should choose a Setup type. It is recommended that you select **Install Client**

Utilities and Driver. Select **Install Driver Only** to install driver only (shown in Figure 2-4). Click **Next** to continue.

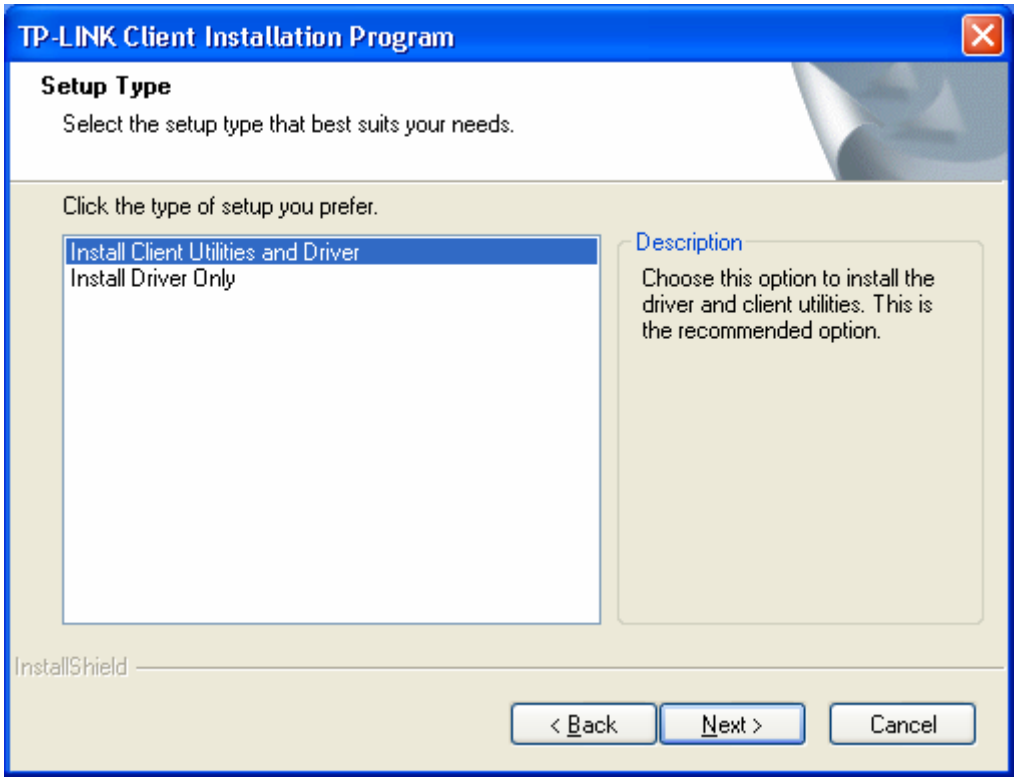


Figure 2-4

5. Click **Browse** to change the destination location for the software, then click **Next** in the screen below (shown in Figure 2-5).

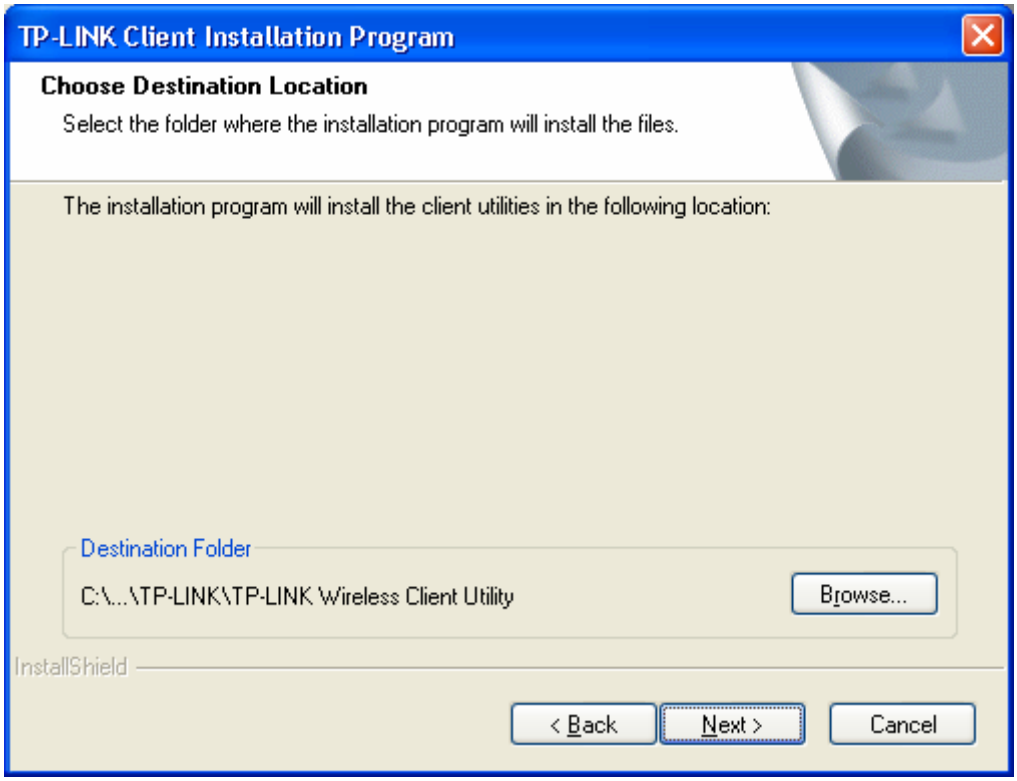


Figure 2-5

6. After that, select the program folder, you should create a new folder name or select one from the **Existing Folders** list. It is recommended that you keep the default setting. Click **Next** to continue the installation.

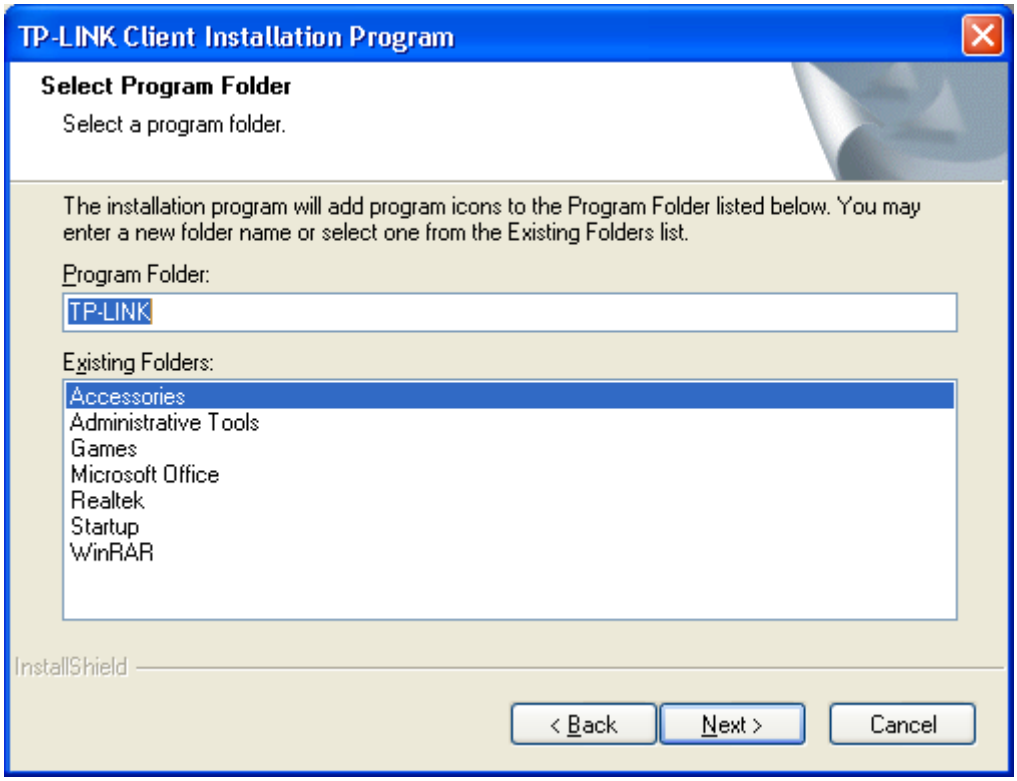


Figure 2-6

7. Choose configuration tool, if you are not sure, please leave it by default. Then click **Next** to continue.

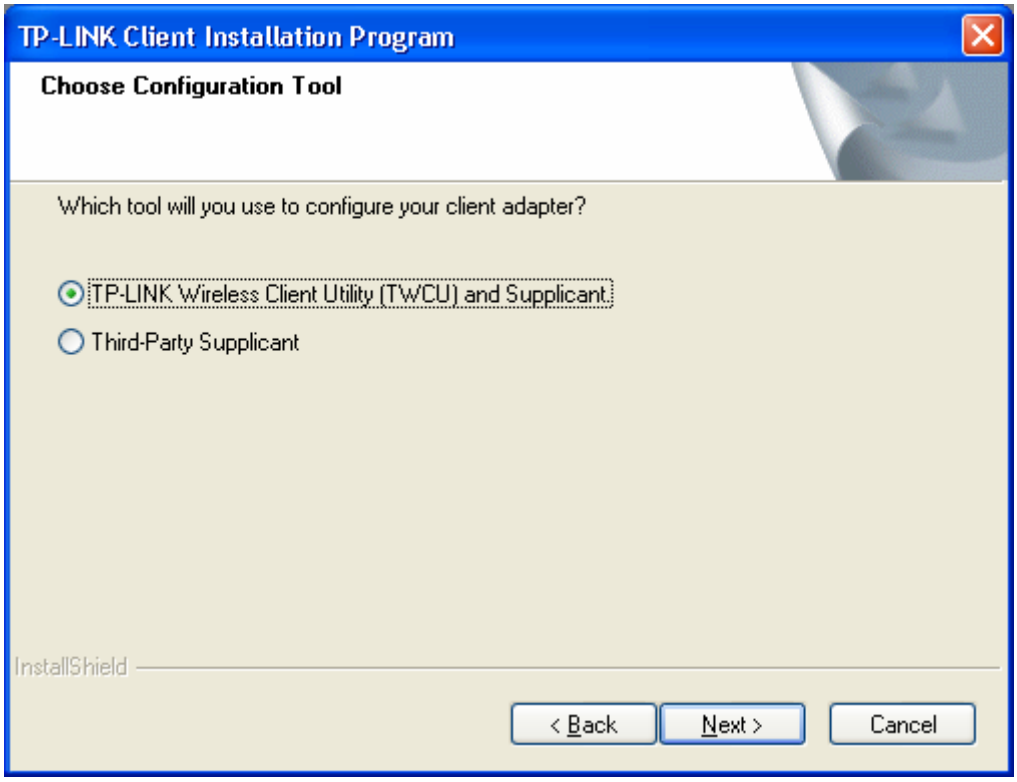


Figure 2-7

8. Wait a while for the setup as shown in Figure 2-8.

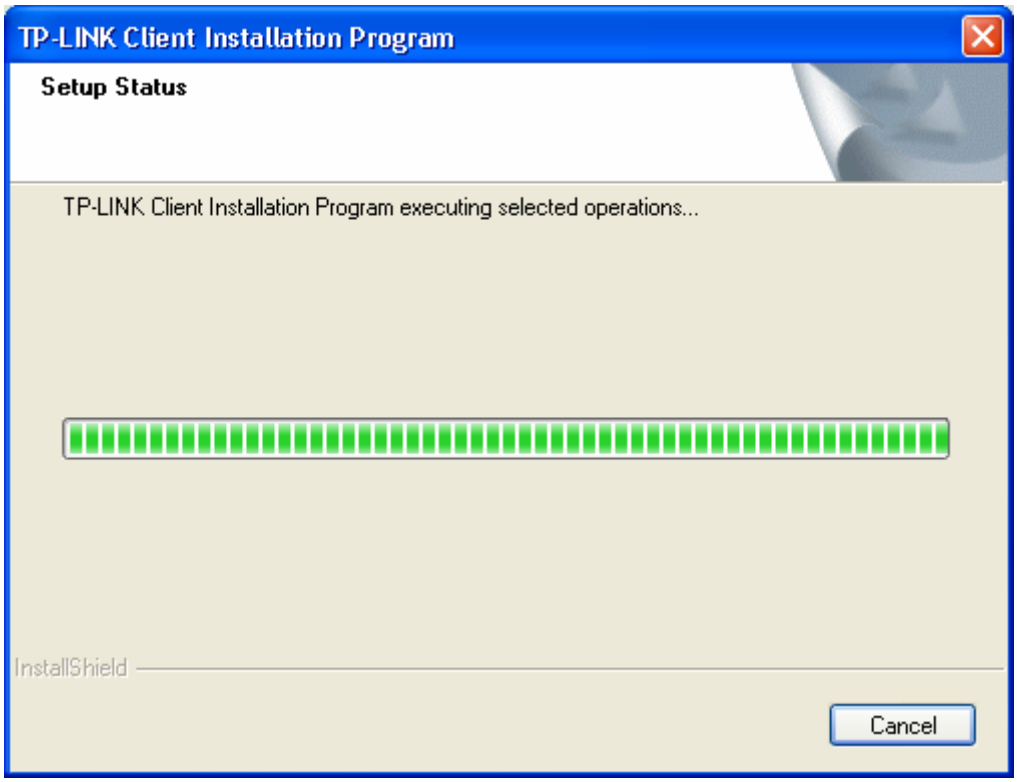


Figure 2-8

9. After all the steps above, you will see the screen below, click **Finish** to reboot the system.

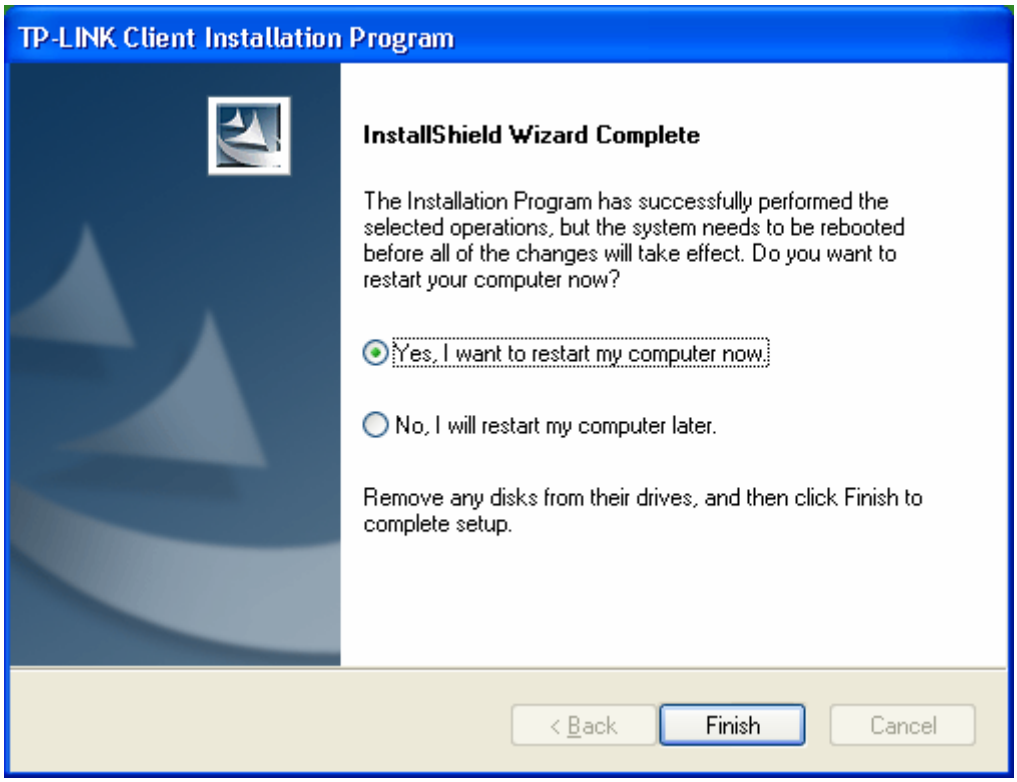


Figure 2-9

2.2.2 For Windows Vista

1. Insert the Resource CD into your CD-ROM drive, and the Figure 2-10 will appear. To continue, select the **TL-WN821N**.There will be a menu including: **Install Driver & Utility**, **Install QSS** and **User Guide**.



Figure 2-10

2. Click **Install Driver & Utility** to load the following screen.

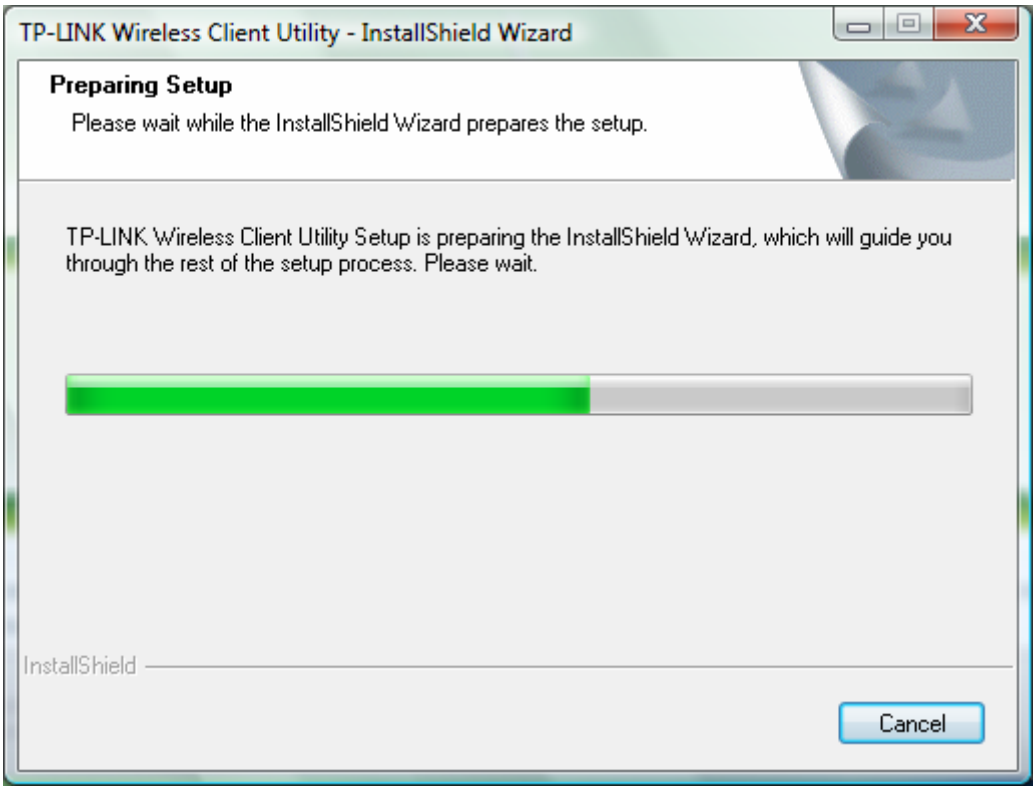


Figure 2-11

3. Soon, Figure 2-12 will display after a moment. Click **Next** to continue.

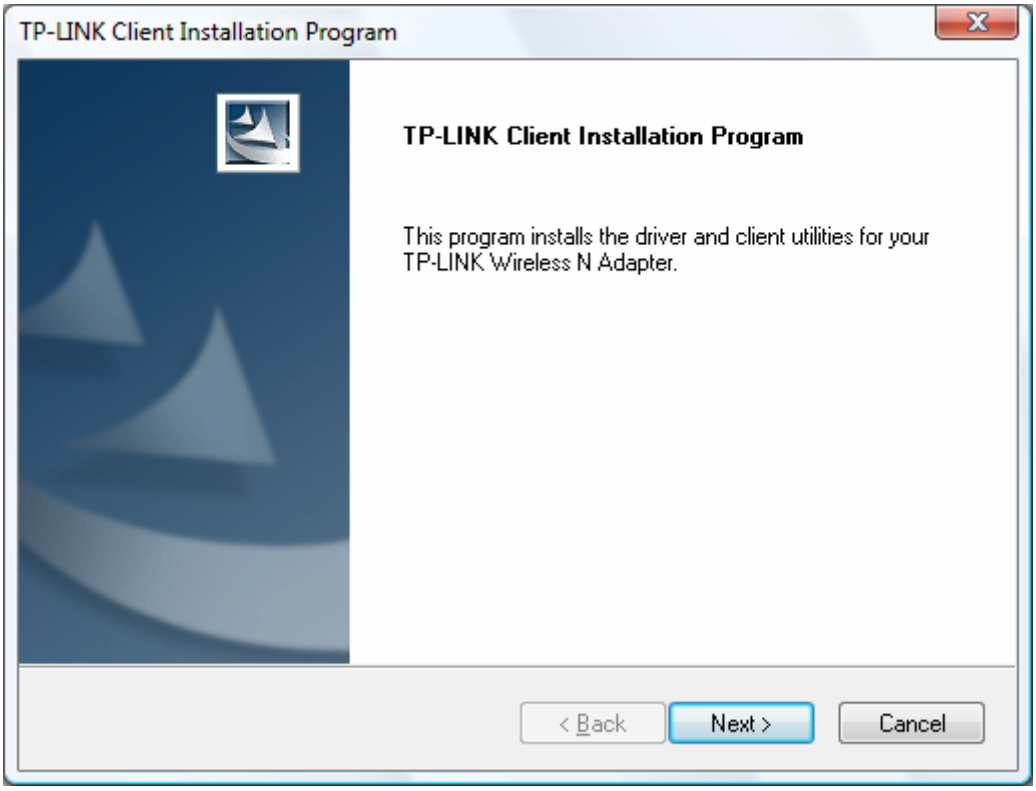


Figure 2-12

4. Click **Yes** to continue.

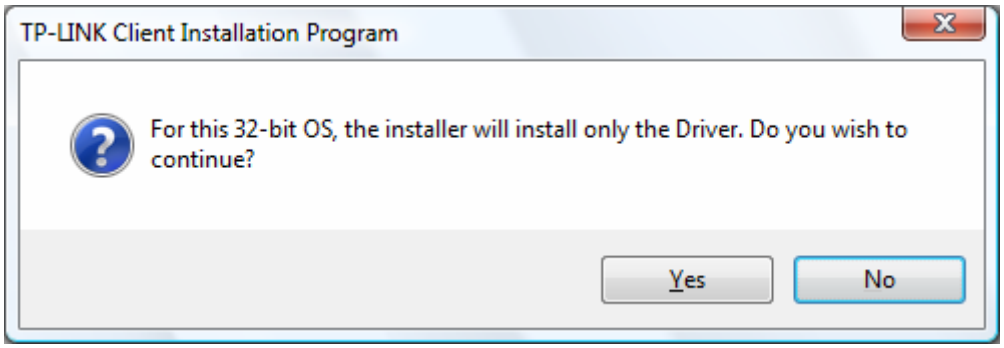


Figure 2-13

5. The following page will display and please wait a moment.

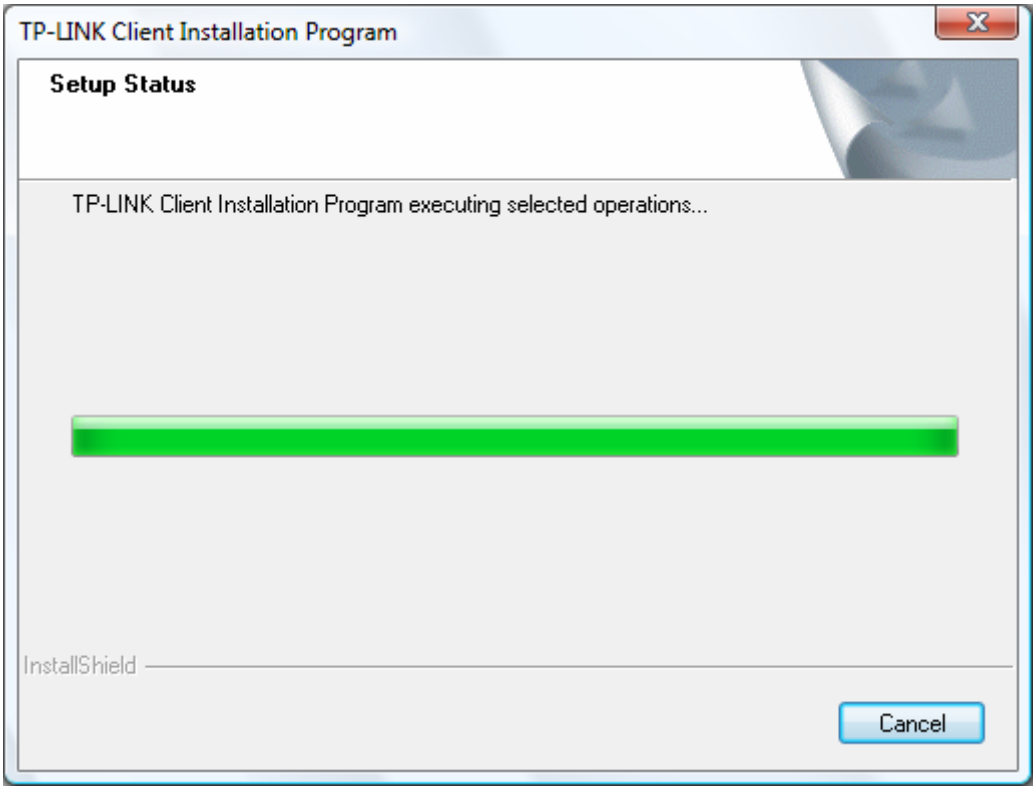


Figure 2-14

6. Click **Finish** to complete the installation.

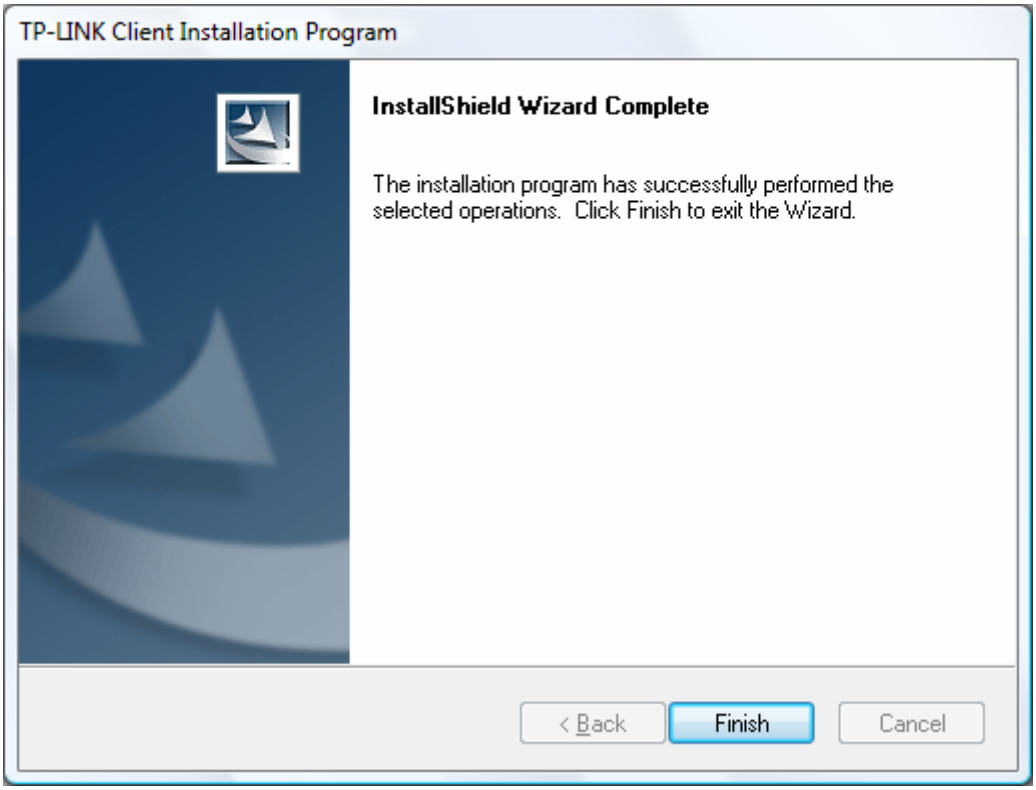


Figure 2-15

2.2.3 For Windows 7

1. Insert the Resource CD into your CD-ROM drive, and the Figure 2-16 will appear. To continue, select the **TL-WN821N**.There will be a menu including: **Install Driver & Utility**, **Install QSS** and **User Guide**.



Figure 2-16

2. Click **Install Driver & Utility** to load the following screen.

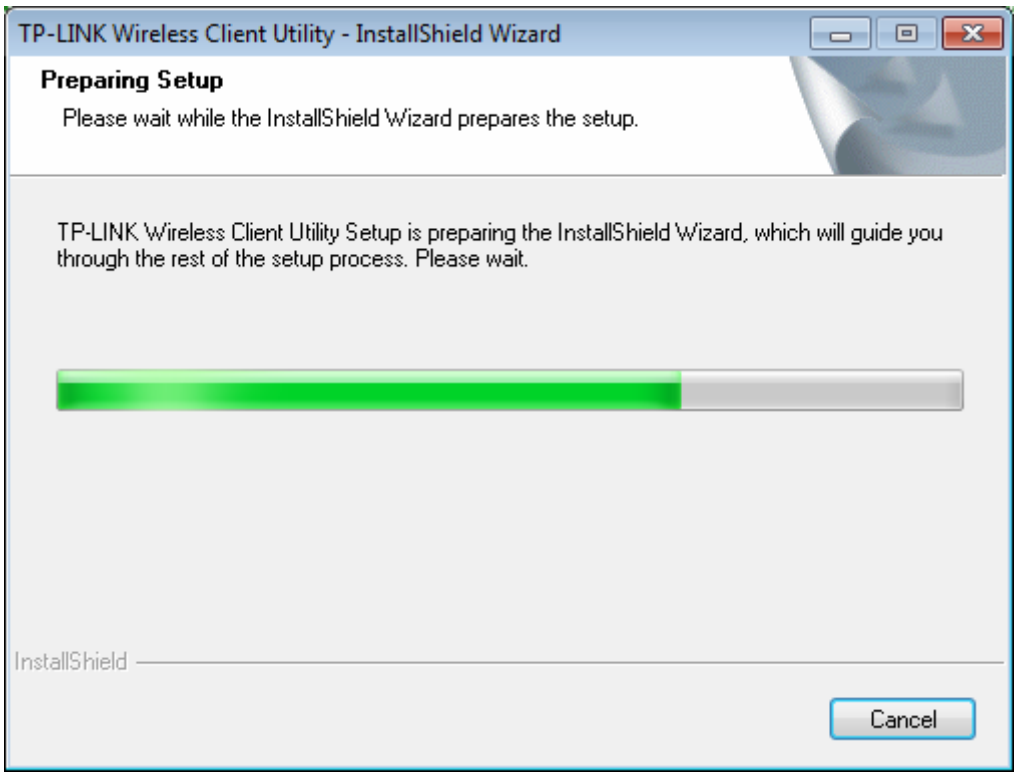


Figure 2-17

3. Soon, Figure 2-18 will display after a moment. Click **Next** to continue.

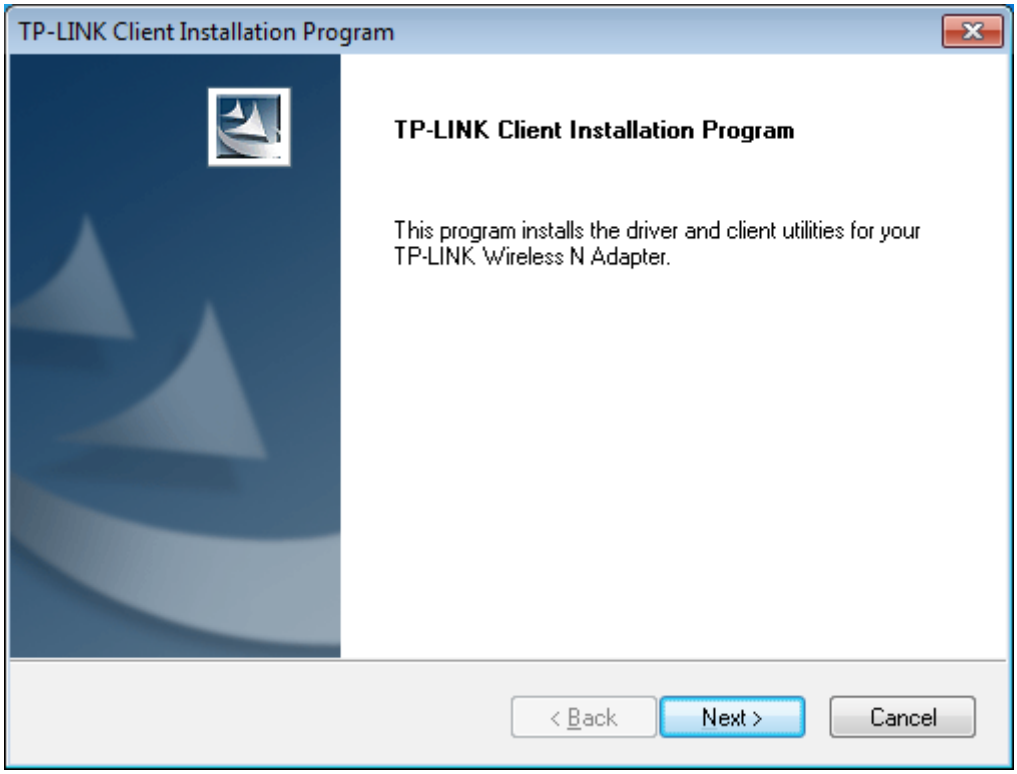


Figure 2-18

4. Click **Yes** to continue.

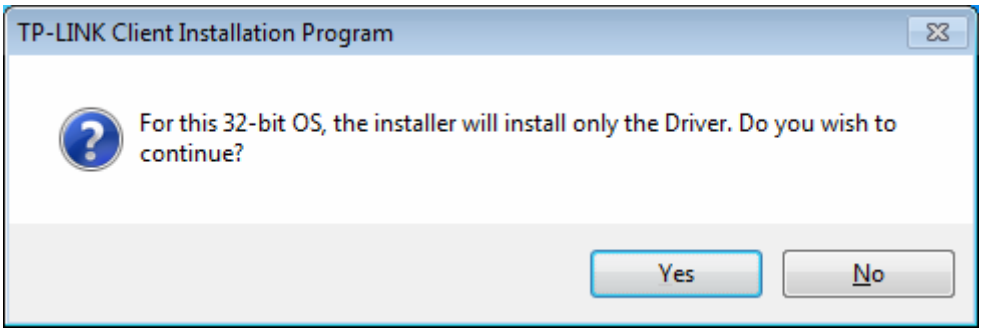


Figure 2-19

5. Wait a while for the setup as shown in Figure 2-20.

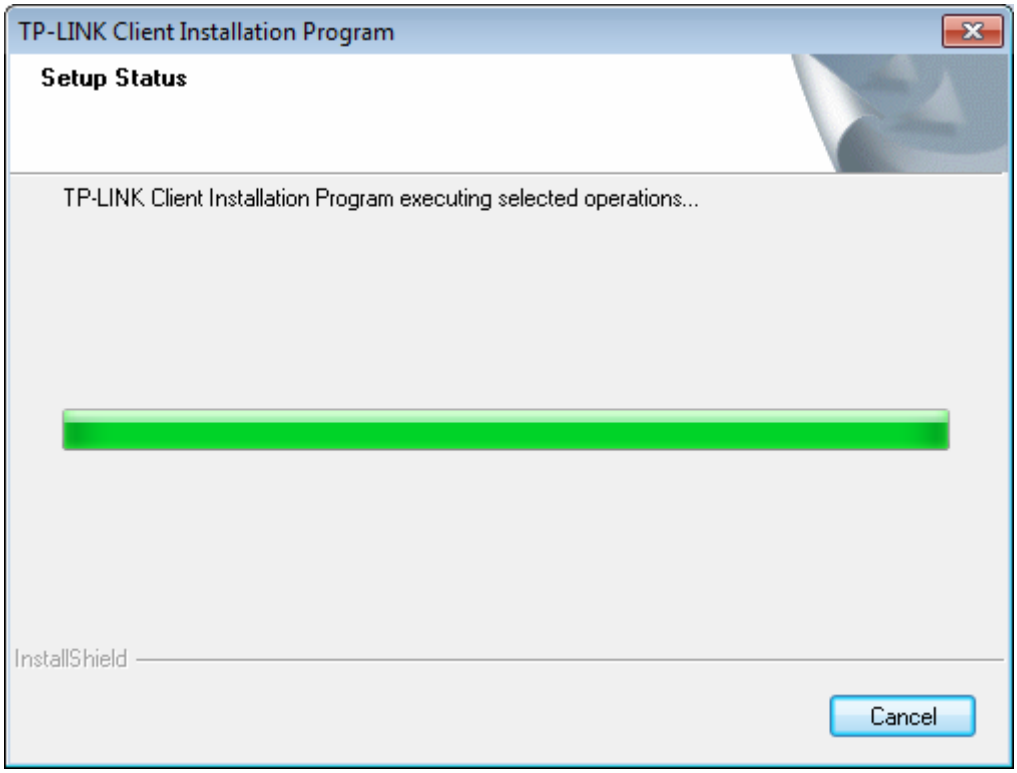


Figure 2-20

6. Then you will see Figure 2-21. Click **Finish** to complete.

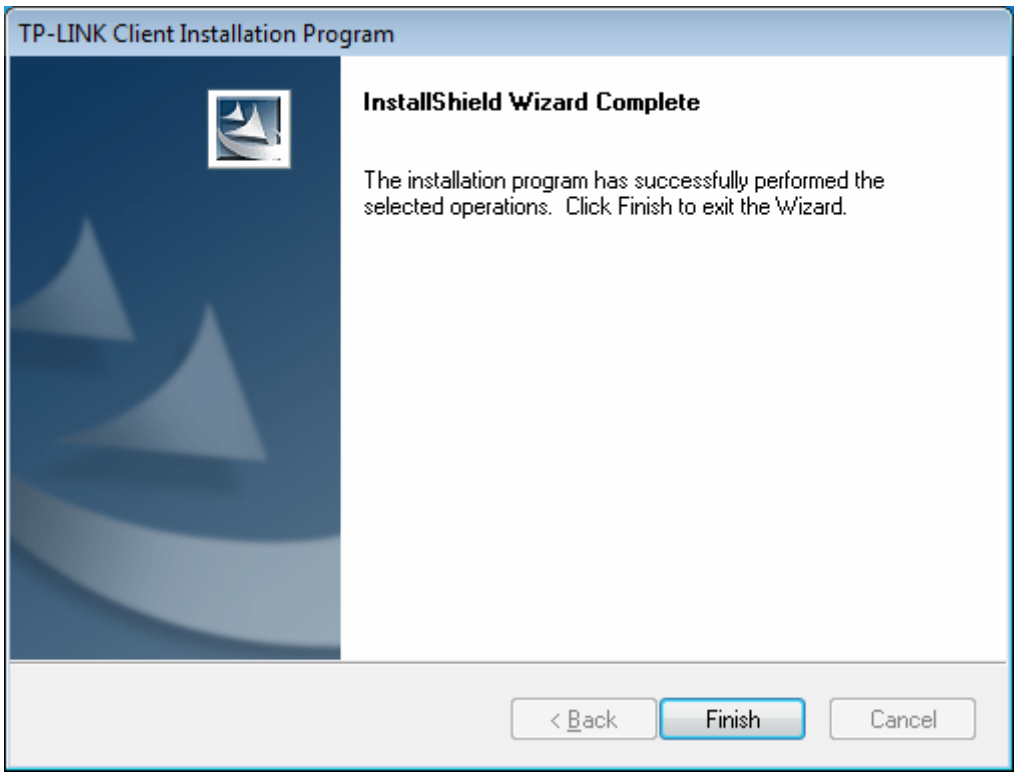

TP-LINK Client Installation Program
InstallShield Wizard Complete
The installation program has successfully performed the selected operations. Click Finish to exit the Wizard.
< Back
Finish
Cancel

Figure 2-21

Chapter 3. Configuration Guide

3.1 For Windows XP

TP-LINK TL-WN821N can be configured by TP-LINK Wireless N Client Utility (TWCU). This chapter describes how to configure your Adapter for wireless connectivity on your Wireless Local Area Network (WLAN) and use the data security encryption features.

After Installing the Adapter, the Adapter's tray icon  will appear in your system tray. It appears at the bottom of the screen, and shows the signal strength using color and the received signal strength indication (RSSI).

 If the icon is gray, there is no connection.

 If the icon is red, there is poor signal strength and the RSSI is less than 5dB.

 If the icon is yellow, there is poor signal strength and the RSSI is between 5dB and 10dB.

 If the icon is green, there is good signal strength and the RSSI is between 10dB and 20dB.

 If the icon is green, there is excellent signal strength and the RSSI is more than 20dB.

Double-click the icon and the **TWCU** will run. You can also run the utility by clicking the **Start** → **All Programs** → **TP-LINK** → **TP-LINK Wireless N Client Utility**. The TWCU provides some integrated and easy tools to:

- Display current status information
- Edit and add configuration profiles
- Display current diagnostics information

The section below introduces these above capabilities.

3.1.1 Current Status

The Current Status tab contains general information about the program and its operations. The Current Status tab needn't any configurations.

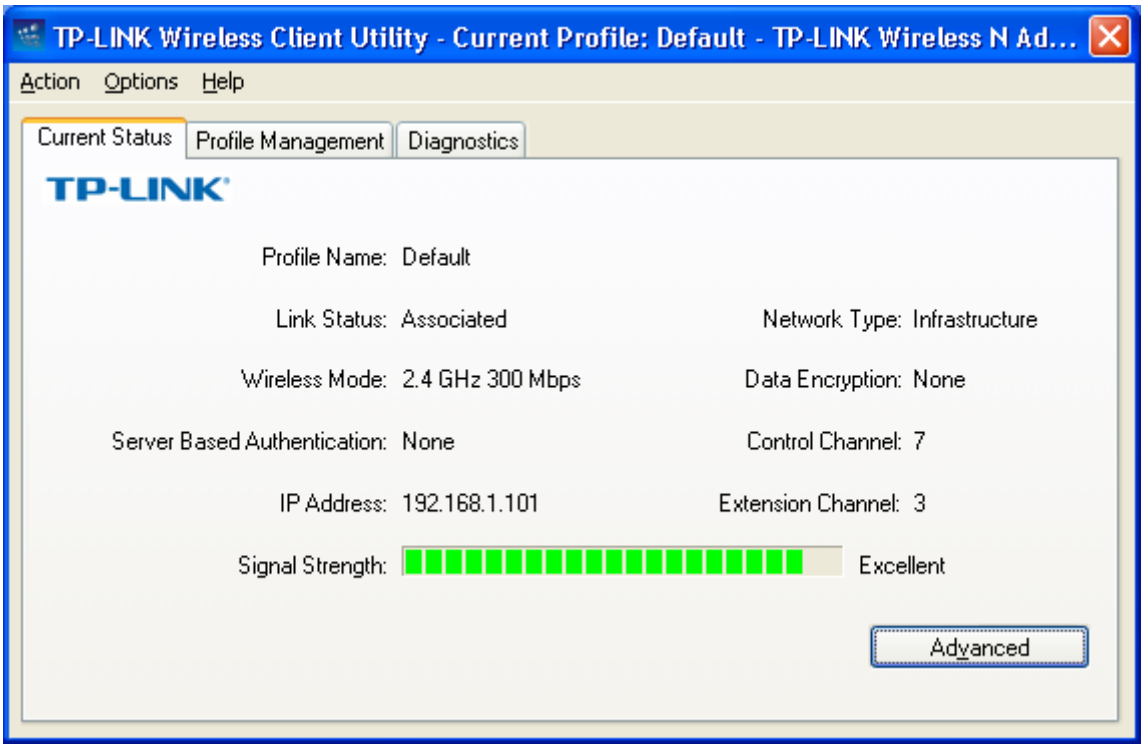


Figure 3-1

The following table describes the items found on the Current Status screen.

- **Profile Name -** This shows the name of current selected configuration profile. The configuration of Profile name will be described on the **General** tab of **Profile Management**.
- **Link Status -** This shows whether the station is associated to the wireless network.
- **Wireless Mode -** Here displays the wireless mode.
- **Network Type -** The type of network and the station currently connected are shown here. The options include:
  - Infrastructure (access point)
  - Ad Hoc

Note:

You can configure the network type and wireless mode on the **Advanced** tab of **Profile Management**.

- **IP Address -** This displays the computer's IP address.
- **Control Channel -** This shows the currently connected channel.
- **Data Encryption -** Here displays the encryption type the driver is using. You can configure it on the **Security** tab of **Profile Management**.
- **Server Based Authentication -** This shows whether the server based authentication is used.

- **Signal Strength -** This shows the strength of the signal.

Click **Advanced** on the screen above, you can see advanced information about the program and its operations.

3.1.2 Profile Management

Click the Profile Management tab of the **TWCU** and the next screen will appear (shown in Figure 3-2). The Profile Management screen provides tools to:

- Add a new profile
- Modify a profile
- Remove a profile
- Activate a Profile
- Import a Profile
- Export a Profile
- Scan Available Networks
- Order profiles

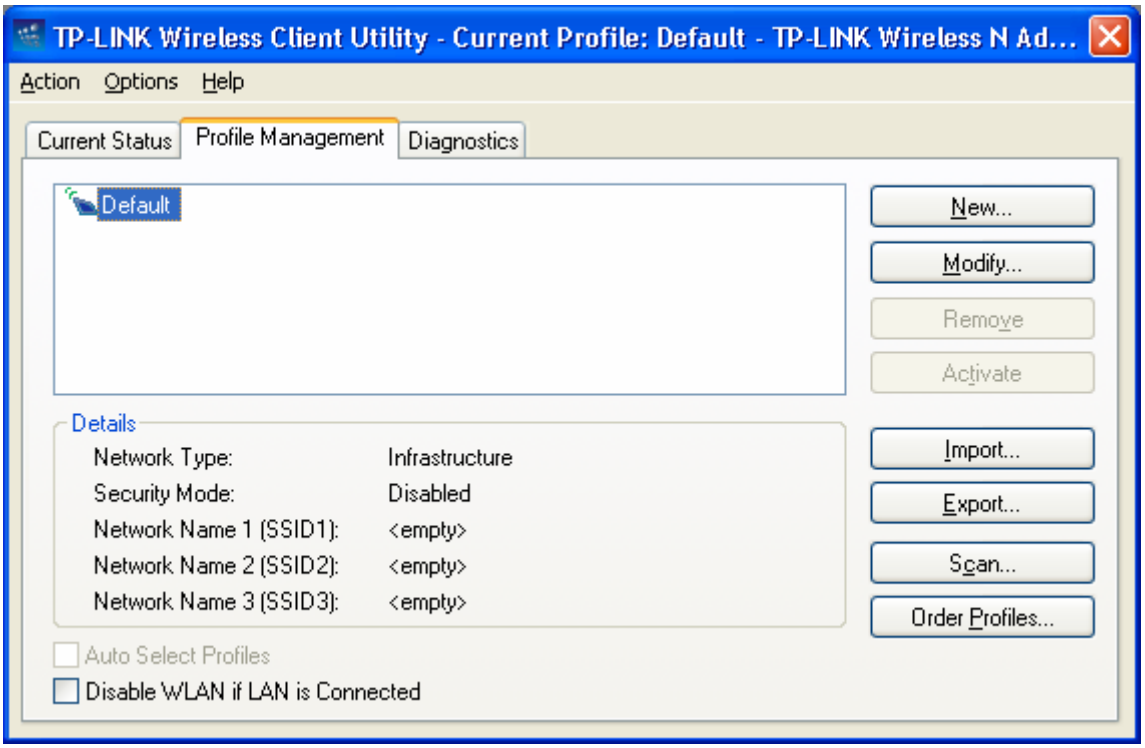


Figure 3-2

3.1.2.1. Add or Modify a Configuration Profile

To add a new configuration profile, click **New** on the Profile Management tab. To modify a configuration profile, select the configuration profile from the Profile list and click **Modify**. Then you will see the Management dialog box (shown in Figure 3-3).

1. Edit the General tab

- **Profile Name -** Please enter the Profile name which identifies the configuration profile. This name must be unique. Note that the profile names are not case-sensitive.
- **Client Name -** Please enter a name which identifies the client machine.
- **Network Names (SSIDs) -** Please enter the IEEE 802.11 wireless network name. This field has a maximum limit of 32 characters.

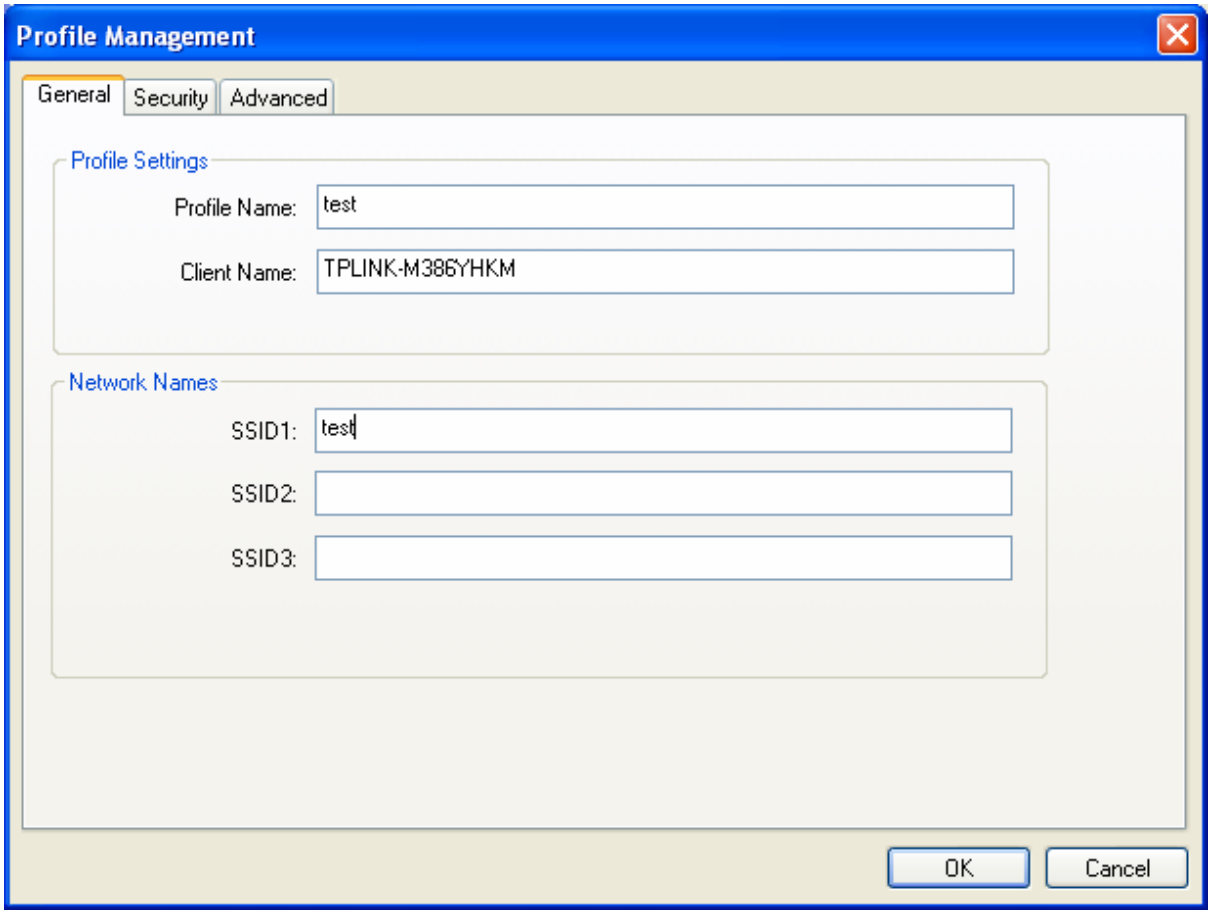


Figure 3-3

2. Edit the Security tab

Select the Security tab in the screen above, and then you can edit the fields to configure the profile. To define the security mode, select the radio button of the desired security mode as follows.

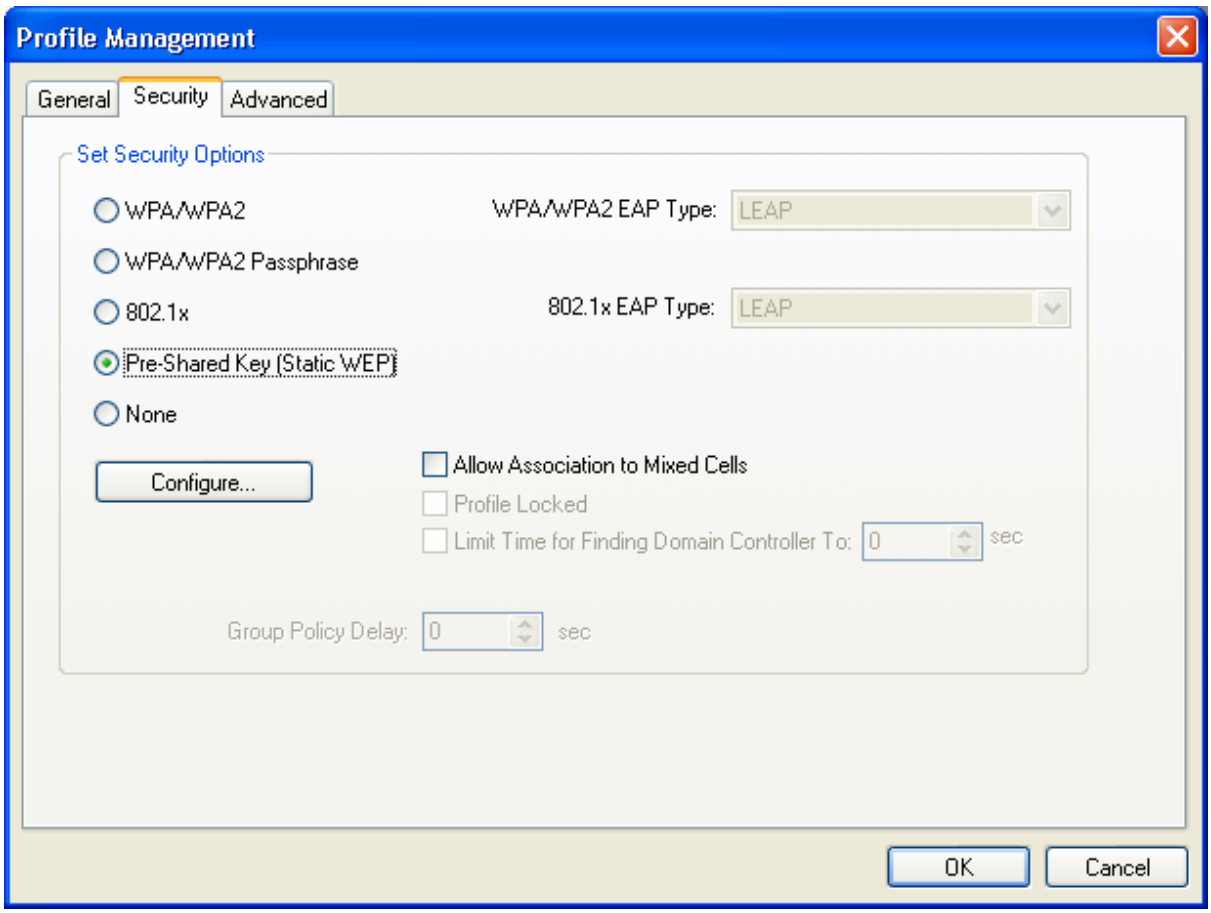


Figure 3-4

- **WPA/WPA2:** Wi-Fi Protected Access
- **WPA/WPA2 Passphrase:** Wi-Fi Protected Access Passphrase
- **802.1x:** Enables 802.1x security.
- **Pre-Shared Key (Static WEP):** Enables the use of shared keys that are defined on both the access point and the station. To define shared encryption keys, choose the Shared Key radio button and click **Configure** to fill in the Define Shared Keys window (shown in Figure 3-5).

Note:

The **WEP** security mode is not available for 802.11n.

- **None:** No security (not recommended).

Note:

If the access point which the Adapter is associated has WEP set and the client has WEP enabled, make sure that **Allow Association to Mixed Cells** is checked on the Security tab to allow association. To complete WEP encryption configuration, you must select the 802.11 Authentication Mode as appropriate on the **Advanced** tab of this **Profile Management** dialog.

To configure the Encryption Keys under the Pre-Shared keys (Static WEP) Security mode:

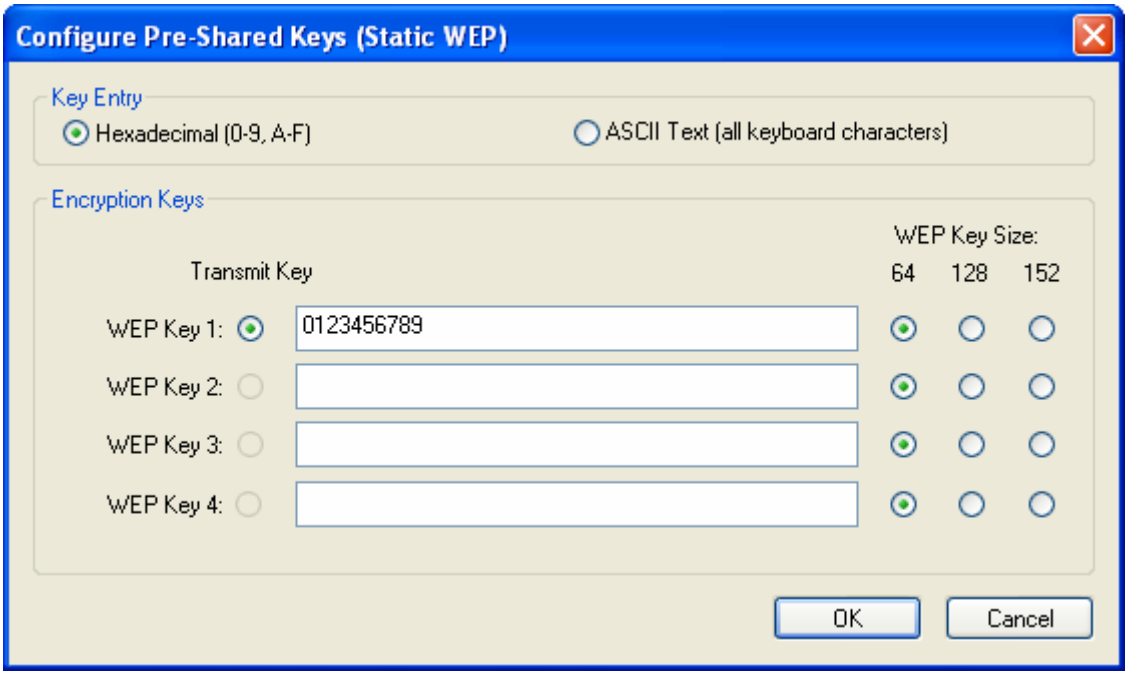


Figure 3-5

Note:

Select different **Security Options**, the configurations are different; you can select the appropriate security option and configure the exact key as your need.

3. Edit the Advanced tab

This screen below allows you to make advanced configuration for the profile.

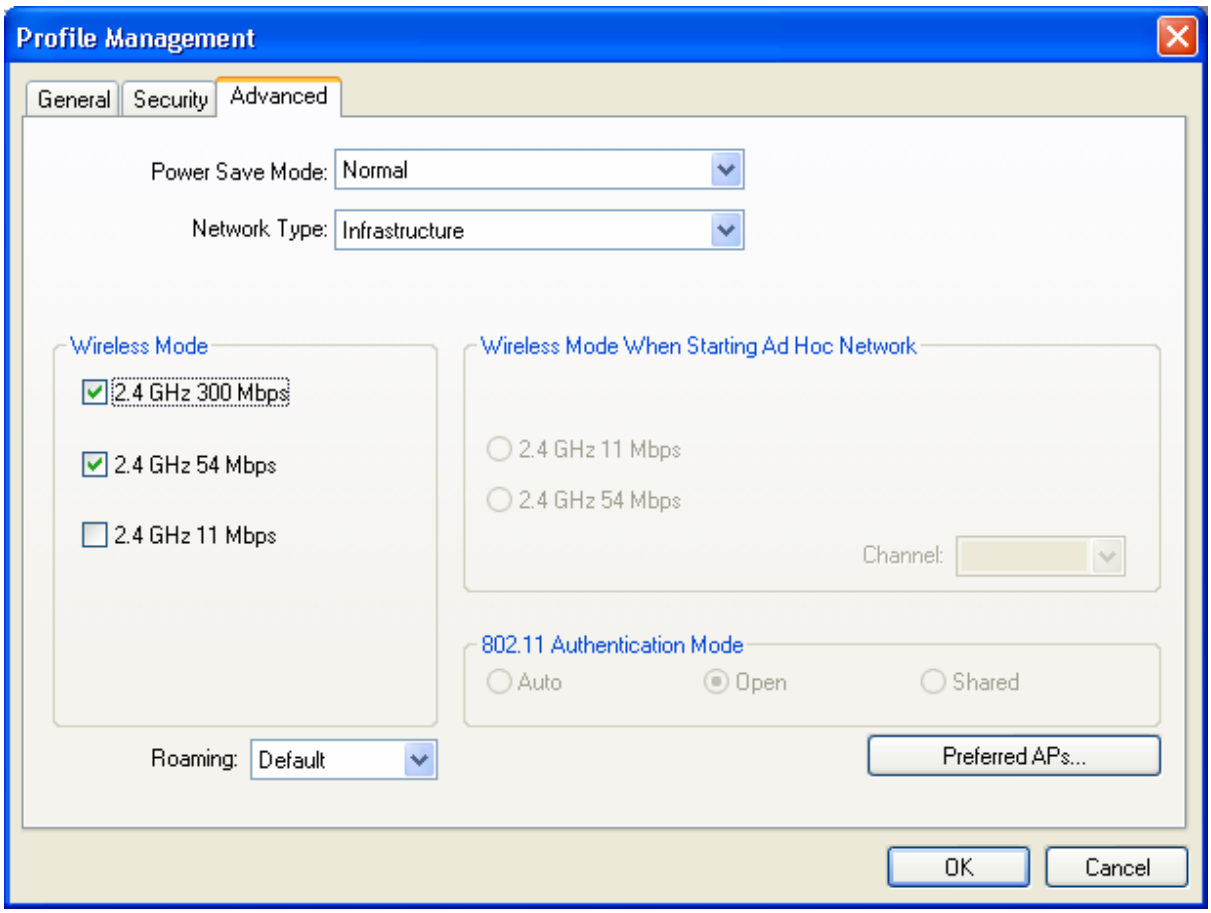


Figure 3-6

- **Power Save Mode -** Please select the power save mode in the drop-down list.
  - **Maximum -** Selects maximum mode to let the access point buffer incoming messages for the Adapter. The Adapter will detect the access point if any messages are waiting periodically.
  - **Normal -** Normal mode uses maximum when retrieving a large number of packets, then switches back to power save mode after retrieving the packets.
  - **Off -** Turns power saving off, thus powering up the Wireless USB Adapter continuously for a short message response time.
- **Network Type:** There are basically two modes of networking:
  - **Infrastructure -** All wireless clients will connect to an access point or wireless router.
  - **Ad-Hoc -** Directly connecting to another computer, for peer-to-peer communication, using wireless network adapters on each computer, such as two or more TL-WN821N wireless adapters.

Note:

1) An Infrastructure network contains an Access Point or wireless router. All the wireless devices or clients will connect to the wireless router or access point.

2) An Ad-Hoc network contains only clients, such as laptops with wireless desktop adapters. All the adapters must be in Ad-Hoc mode to communicate.

- **Wireless Mode**: Specifies 2.4 GHz 150 Mbps, 2.4 GHz 54 Mbps or 2.4 GHz 11 Mbps operation in an access point network. The Wireless adapter must match the wireless mode of the access point with which it associates.

- **Wireless Mode when Starting an Ad Hoc Network**: Specifies 2.4 GHz 54/11 Mbps to start an Ad Hoc network if no matching network name is found after scanning all available modes. This mode also allows the selection of the channel that the Wireless Adapter uses. The channels available depend on the regulatory domain. If the adapter finds no other ad hoc adapters, the channel that the adapter starts the ad hoc network with will be selected automatically. The Adapter must match the wireless mode and channel of the clients it associates.

- **802.11 Authentication Mode**: Select which mode the Adapter uses to authenticate to an access point:

  - **Auto -** Automatic causes the adapter to attempt authentication using shared, but switches it to open authentication if shared fails.
  - **Open -** Open System enables an adapter to attempt authentication regardless of its WEP settings. It will only associate with the access point if the WEP keys on both the adapter and the access point match.
  - **Shared -** Shared-key only allows the adapter to associate with access points that have the same WEP key.

For infrastructure (access point) networks, click **Preferred APs…** to specify four access points at most to the client adapter that attempts to be associated to the access points. The four access points have different priorities; the frontal has the higher priority.

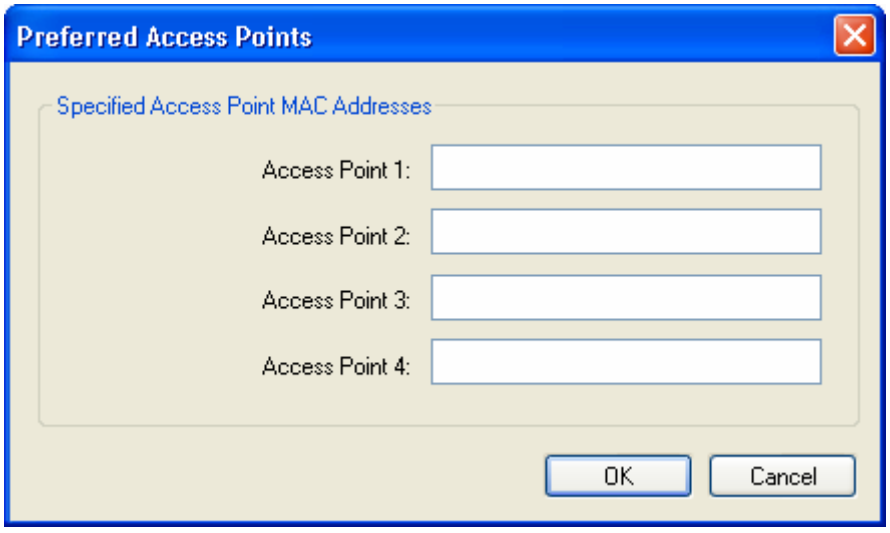


Figure 3-7

3.1.2.2. Remove a profile

1. Go to the Profile Management tab (shown in Figure 3-2).
2. Select the profile name in the Profiles List.
3. Click **Remove**.

Note:

The profile being used can't be removed.

3.1.2.3. Switch another Profile

1. Go to the Profile Management screen (shown in Figure 3-2).
2. Select the profile name required in the Profiles List.
3. Click **Activate**.

3.1.2.4. Export a Profile

1. From the Profile Management screen (shown in Figure 3-2), highlight the profile to export.
2. Click **Export…**, the Export Profile window will then appear below.
3. Browse the directory to export the profile to.
4. Click **Save**. The profile should then be exported to the specified location.

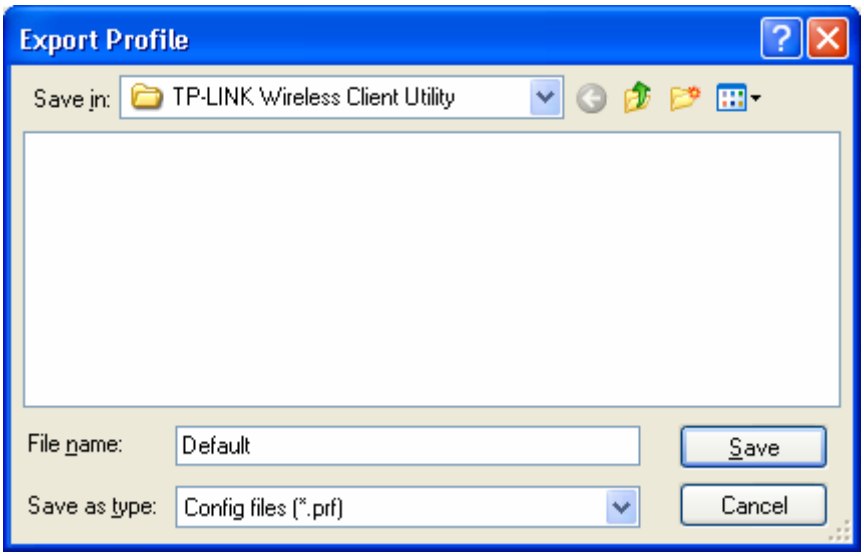


Figure 3-8

3.1.2.5. Import a Profile

1. From the Profile Management screen (shown in Figure 3-2), click **Import…**. Then the Import Profile will appear below.
2. Browse to the directory where the profile is located.

3. Highlight the profile name.

4. Click **Open**, the imported profile will then appear in the Profiles List.

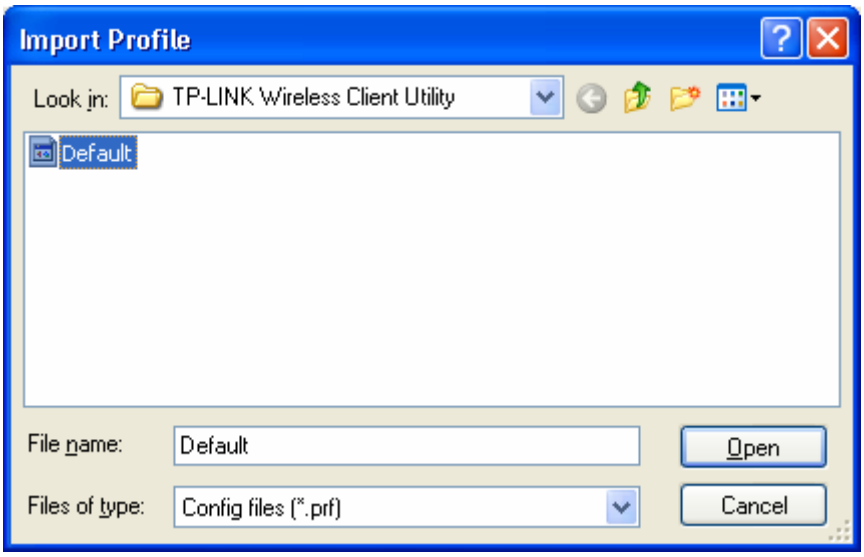


Figure 3-9

3.1.2.6. Scan Available Networks

1. Click **Scan** on the Profile Management screen (shown in Figure 3-2), the Available Infrastructure and Ad Hoc Networks window will appear below.

2. Click **Refresh** to refresh the list at any time.

3. Highlight a network name and click **Activate** to connect to an available network. If no configuration profile exists for that network, the Profile Management window will open the **General** tab screen. Fill in the Profile name and click **OK** to create the configuration profile for that network.

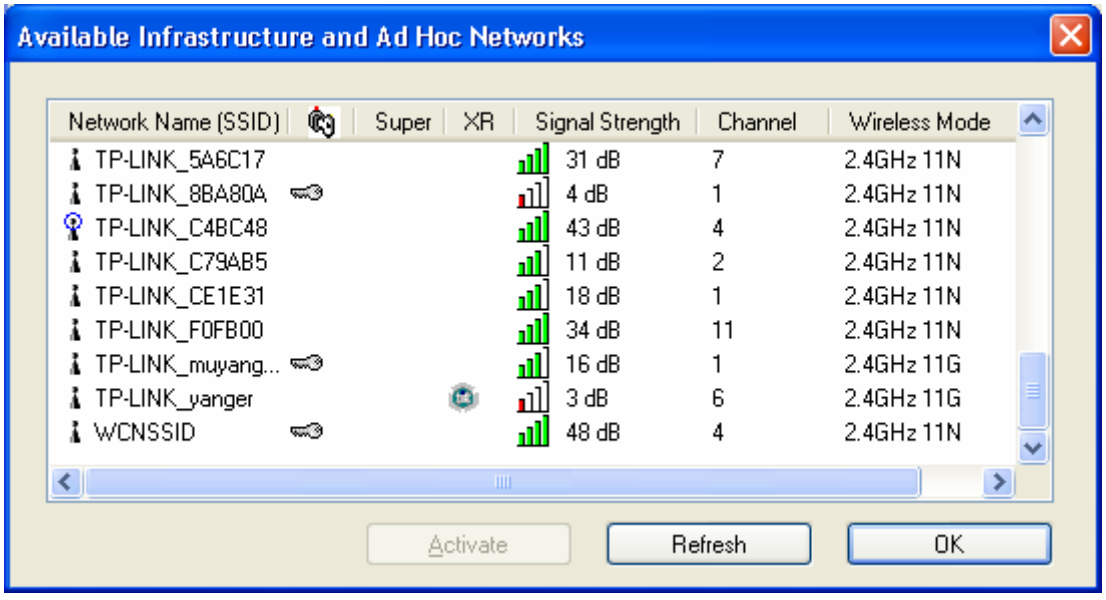


Figure 3-10

3.1.2.7. Auto Profile Selection Management

The auto selection feature allows the adapter to automatically select a profile from the list of profiles and use it to connect to the network. To add a new profile into the Auto Selected Profiles list, please follow these steps.

1. On the Profile Management screen (shown in Figure 3-2), click **Order Profiles…**.
2. The Auto Profiles Selection management window will appear (shown in Figure 3-11) with a list of all created profiles in the Available Profiles.

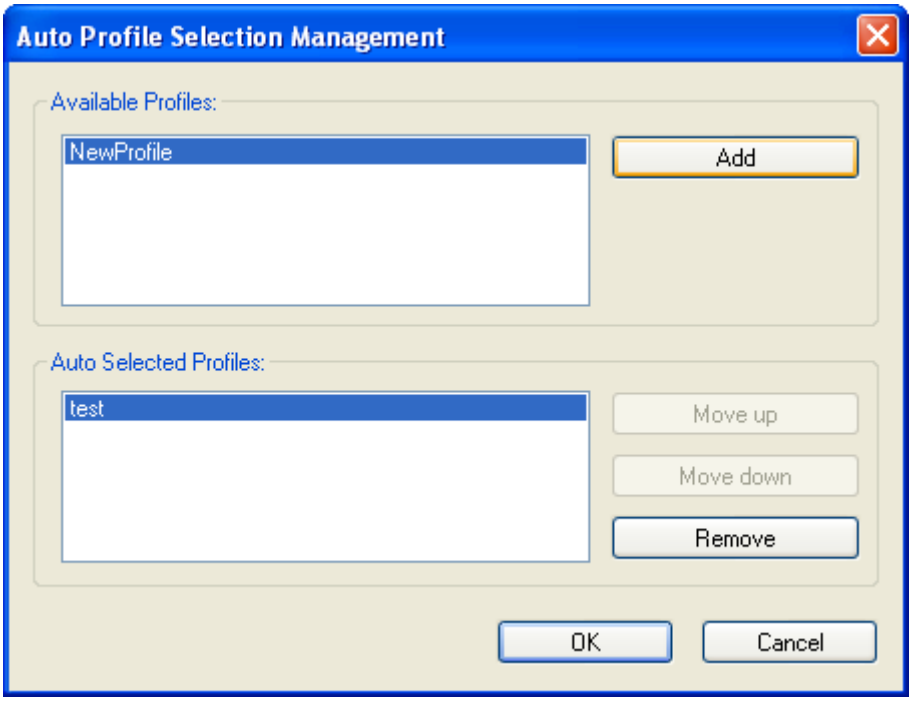


Figure 3-11

3. Highlight the profiles to add to auto profile selection, and click **Add**. The profile will appear in the Auto Selected Profiles box.
4. Highlight a profile in the Auto Selected Profiles box.
5. Click **Move Up** or **Move Down** as appropriate.

☞ **Note:**

The first profile in the Auto Selected Profiles box has highest priority, while the last profile has the lowest priority.

6. Click **OK**.
7. Check the **Auto Select Profiles** checkbox on the **Profile Management** tab (shown in Figure 3-2).

☞ **Note:**

When auto profile selection is enabled by checking **Auto Select Profiles** on the **Profile Management** tab, the client adapter will scan for an available network. The profile with the

highest priority and the same SSID as one of the found networks will be used to connect to the network. If the connection fails, the client adapter will try the next highest priority profile that matches the SSID until an available network is found.

3.1.3 Diagnostics

The **Diagnostics** tab of the TP-LINK Wireless N Client Utility (TWCU) provides buttons used to retrieve receiving and transmitting statistics. The Diagnostics tab does not require any configuration.

The Diagnostics tab lists the following receiving and transmitting diagnostics for frames received or transmitted by the wireless network adapter:

- Multicast frames transmitted and received
- Broadcast frames transmitted and received
- Unicast frames transmitted and received
- Total bytes transmitted and received

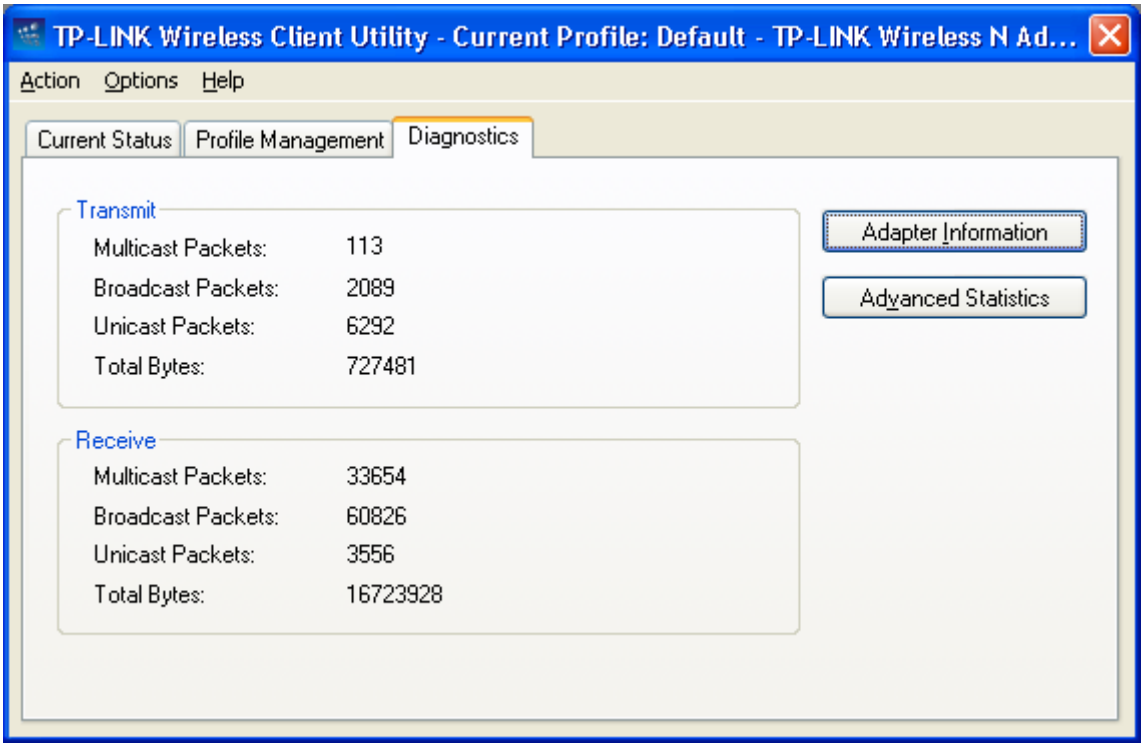


Figure 3-12

3.1.3.1. Check Driver Information

Click the **Adapter Information** button in the screen above, you will see the adapter information, including general information about the wireless network adapter and the Network Driver Interface Specification (NDIS) driver. Access the adapter information from the Diagnostics tab.

- **Card Name -** The name of the wireless network adapter.
- **MAC Address -** The MAC address of the wireless network adapter.

- **Driver -** The driver name and path of the wireless network adapter driver.
- **Driver Version -** The version of the wireless network adapter driver.
- **Driver Date -** The creation date of the wireless network adapter driver.
- **Client Name -** The name of the client computer.

3.1.3.2. Check Receive and Transmit Statistical Information

The **Advanced Statistics** show receiving and transmitting statistical information about the following receiving and transmitting diagnostics for frames received by or transmitted to the wireless network adapter.

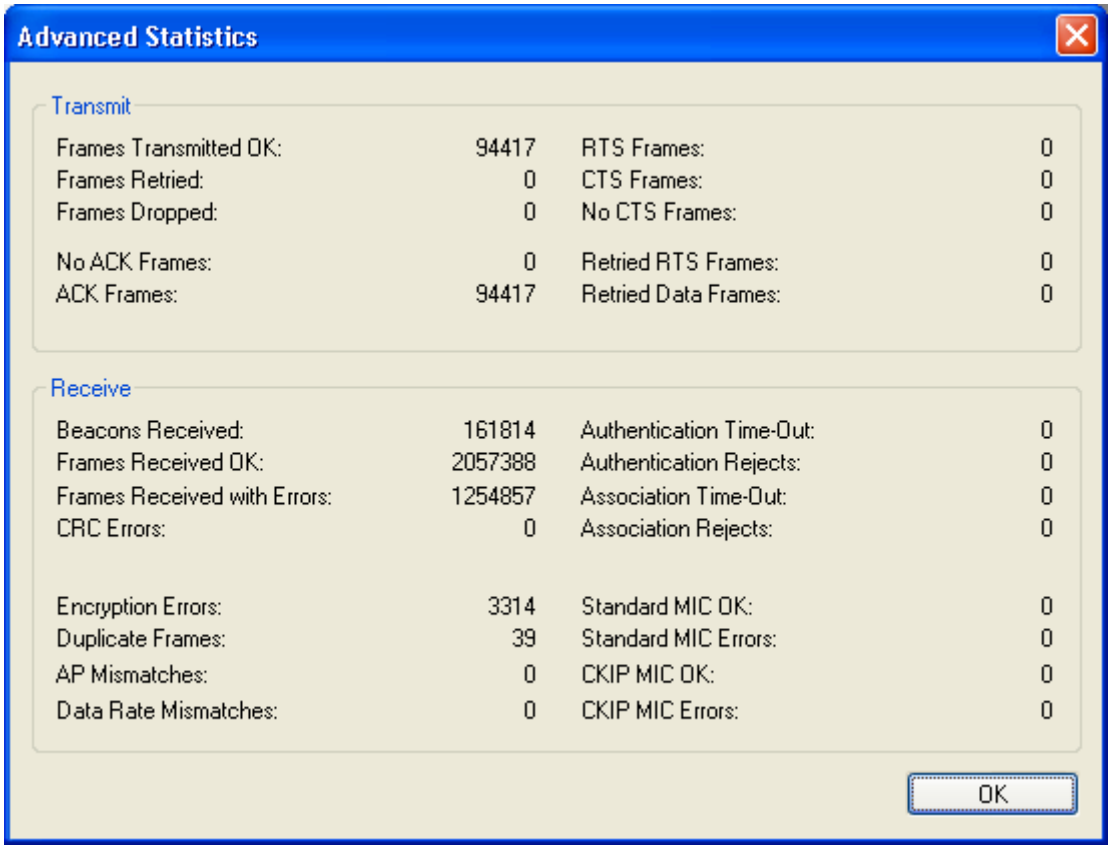


Figure 3-13

3.2 For Windows Vista

After the Adapter's driver has been installed, Windows Vista will display a wireless Network Connection message like this one.



Figure 3-14

Icon  means the connection has been established. Icon  means there is no connection.

To establish a connection, please follow the steps below.

1. Right-click the icon  in your system tray, then click **Connect to a network**.

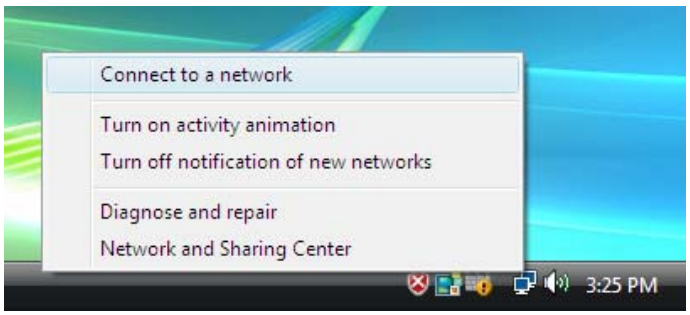


Figure 3-15

2. The following screen will show you available wireless networks. Highlight the one you want to join, and then click **Connect**.

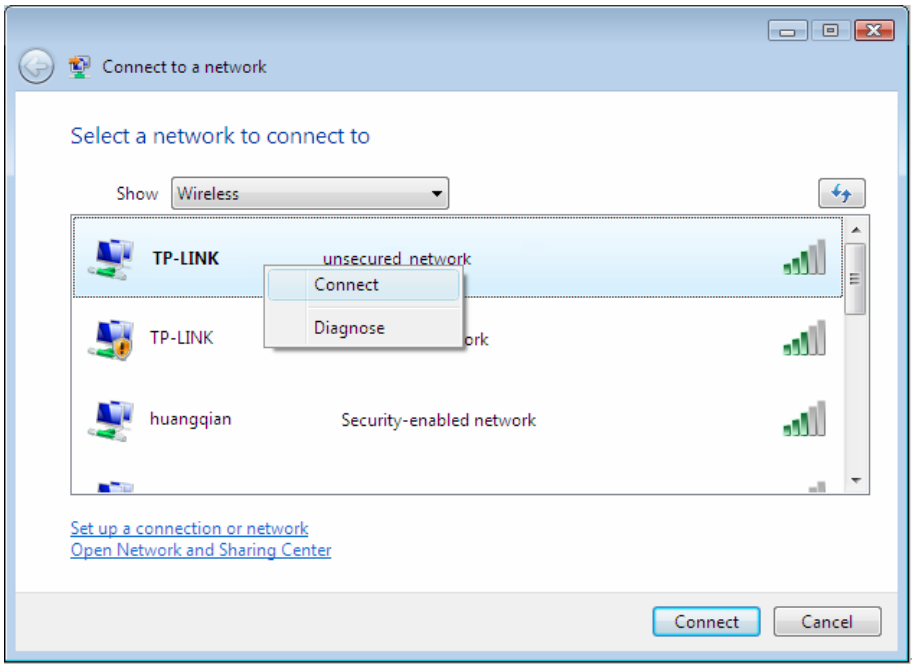


Figure 3-16

3. To continue, click **Connect Anyway**. Click the **Cancel** button to end the connection.

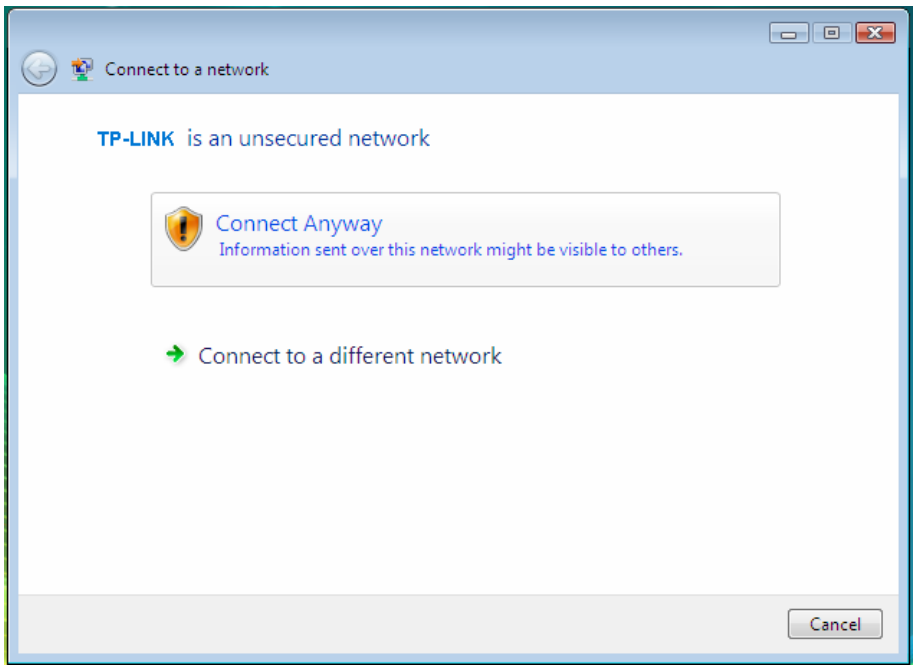


Figure 3-17

4. If the connection is successful established, the following screen will appear, click **close** to finish the connection.

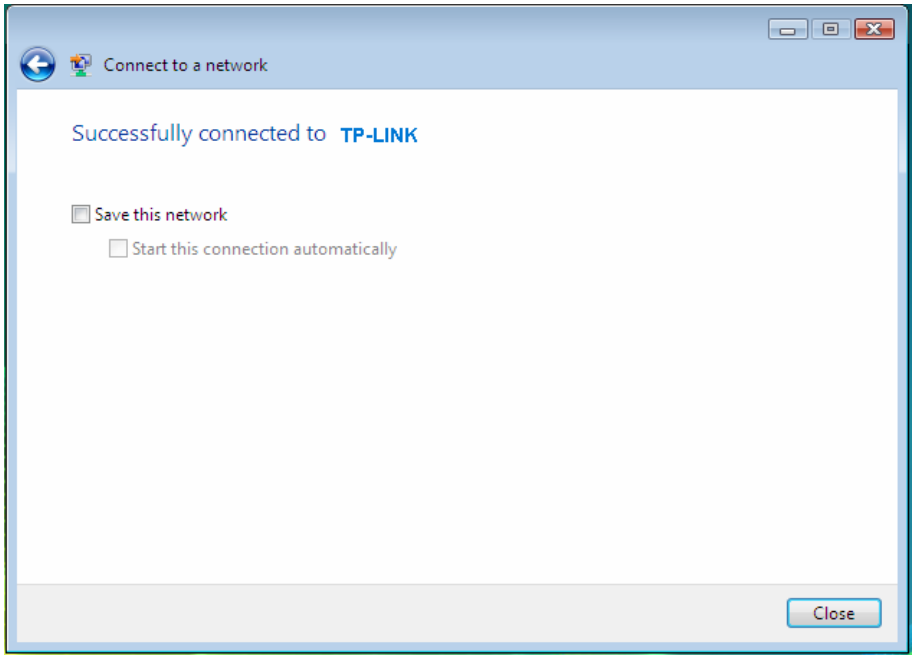


Figure 3-18

3.3 For Windows 7

TP-LINK Wireless N Client Utility (TWCU) is not available for Windows 7. So after the Adapter's driver has been installed, we have to use Windows WLAN Autoconfig to establish a connection. Please follow the steps below.

1. Click the icon  at the bottom of your screen in your system tray and then you will see the available wireless network list. Select the SSID of your Access Point and click **Connect**.

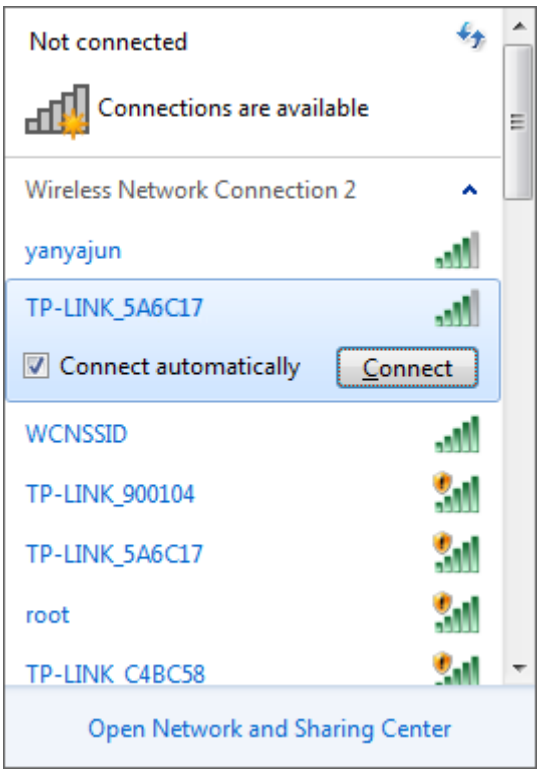


Figure 3-19

2. If your wireless network is secured, you will be required to enter the security key as shown in Figure 3-20. Enter the passphrase, take **1234567890** for example. And then click **OK**.

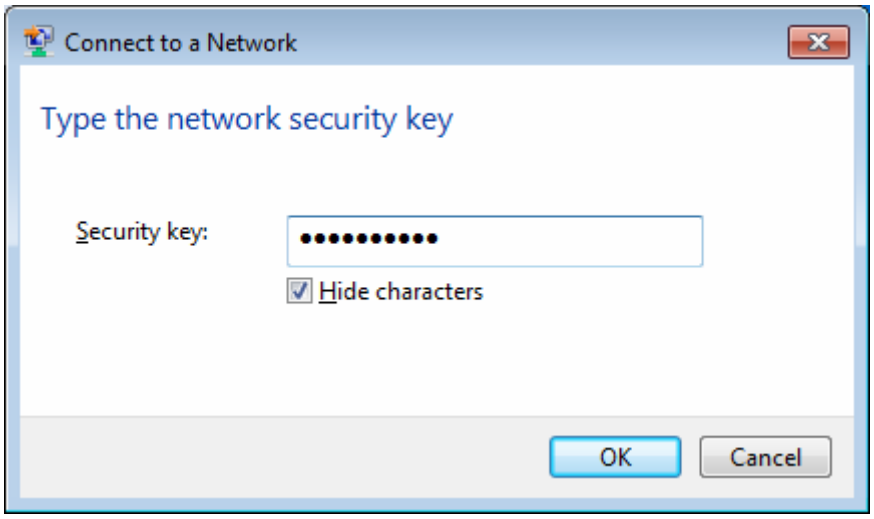


Figure 3-20

3. If the key entered is correct, you will successfully connect to the network as shown in Figure 3-21.


Currently connected to:
TP-LINK_5A6C17
No Internet access
Wireless Network Connection 2
TP-LINK_5A6C17
Connected
Disconnect
WCNSSID
yanyajun
Network-940c6df0fb00
root
TP-LINK_900104
RT3052_WPA_test
Open Network and Sharing Center

Figure 3-21

Chapter 4. QSS configuration

QSS (Quick Secure Setup) function allows you to add a new wireless device to an existing network quickly.

If the wireless card supports QSS (Quick Secure Setup) or Wi-Fi Protected Setup (WPS), you can establish a wireless connection between wireless card and router using either Push Button Configuration (PBC) method or PIN method.

Note:

QSS function is not supported in Windows 7 OS.

First, the QSS software should be installed. Insert the Resource CD into your CD-ROM drive, and the Figure 4-1 will appear. To continue, select the **TL-WN821N**.There will be a menu including: **Install Driver & Utility**, **Install QSS** and **User Guide**.



Figure 4-1

Next, select **Install QSS** to load the following screen.

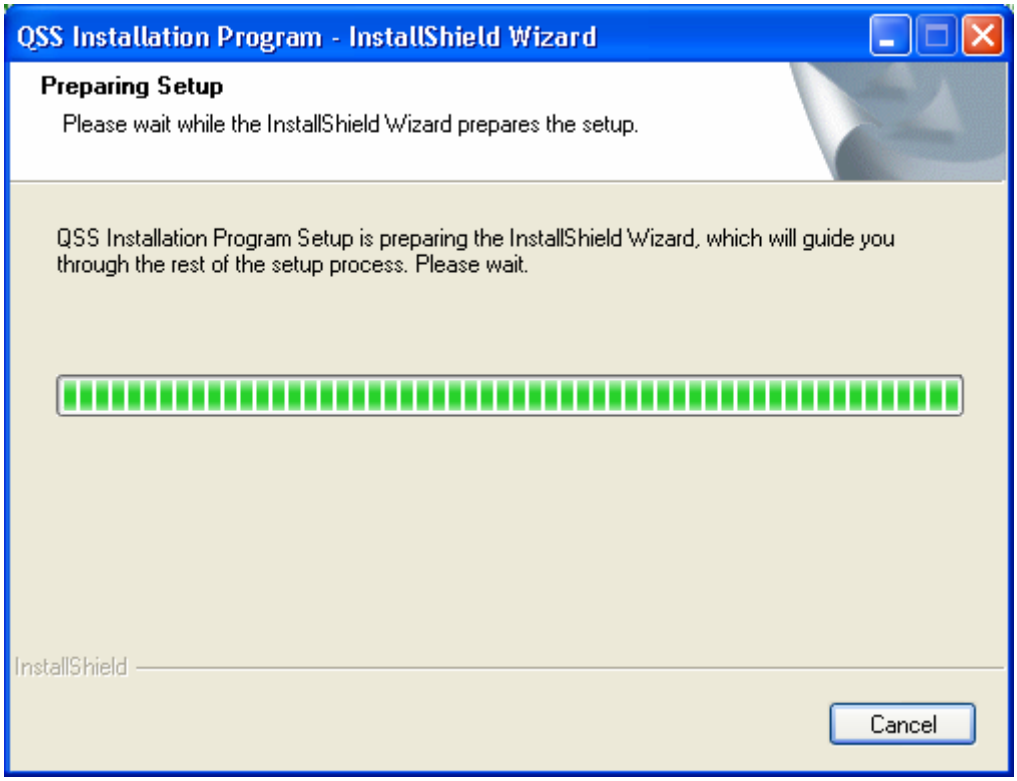


Figure 4-2

Then, please follow the clue of the Operation System to complete the QSS installation. After that, QSS function can be enabled. Here we will introduce two ways to configure the QSS (For the configuration of QSS, here takes the Wireless Router of our company for example).

4.1 PBC (Push Button Configuration) method

1. Firstly, press the QSS button of the Router. There are two ways:

   1) Press the QSS button directly on the front panel of the Router shown in Figure 4-3.



Figure 4-3

   2) Open the Router's Web-based Utility and click **QSS** link on the left of the main menu. Then Figure 4-4 will appear. Click **Add device**, then you can see Figure 4-5. Select **Press the button of the new device in two minutes** and click **Connect**.



Figure 4-4

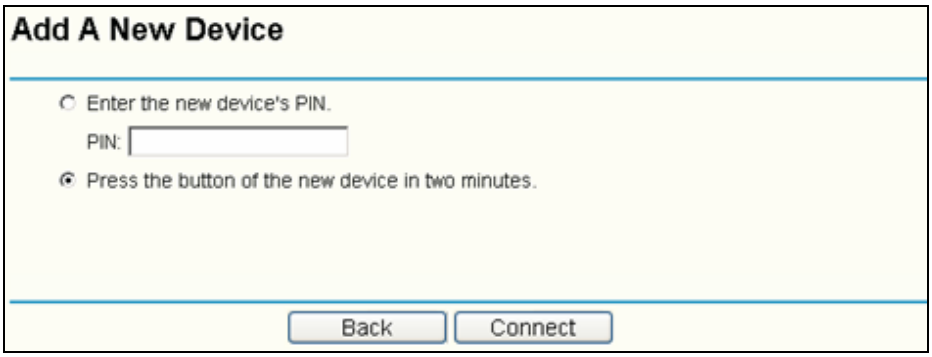


Figure 4-5

2. Secondly, press the QSS button of the adapter. There are two ways:

   1) Press and hold the QSS button of the adapter directly for 2 or 3 seconds.



Figure 4-6

   2) Double click the icon  on the desktop to open the QSS Utility and then you can see the welcome screen shown as Figure 4-7. Click **Next** to continue**.** Then select **Push the button on my access point** in the next screen shown in Figure 4-8 and click **Next**.

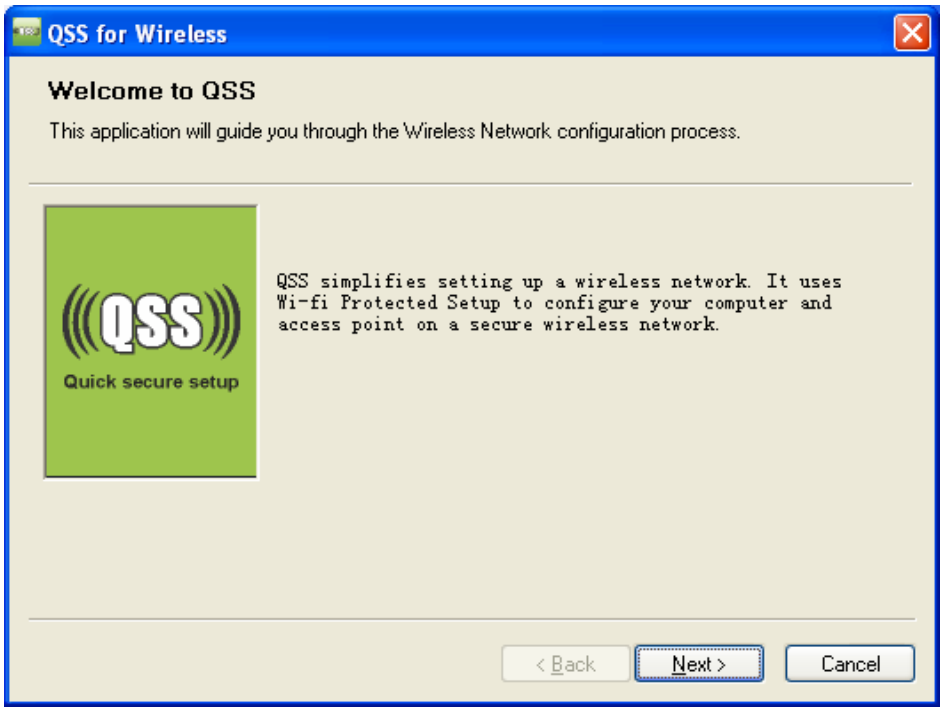


Figure 4-7

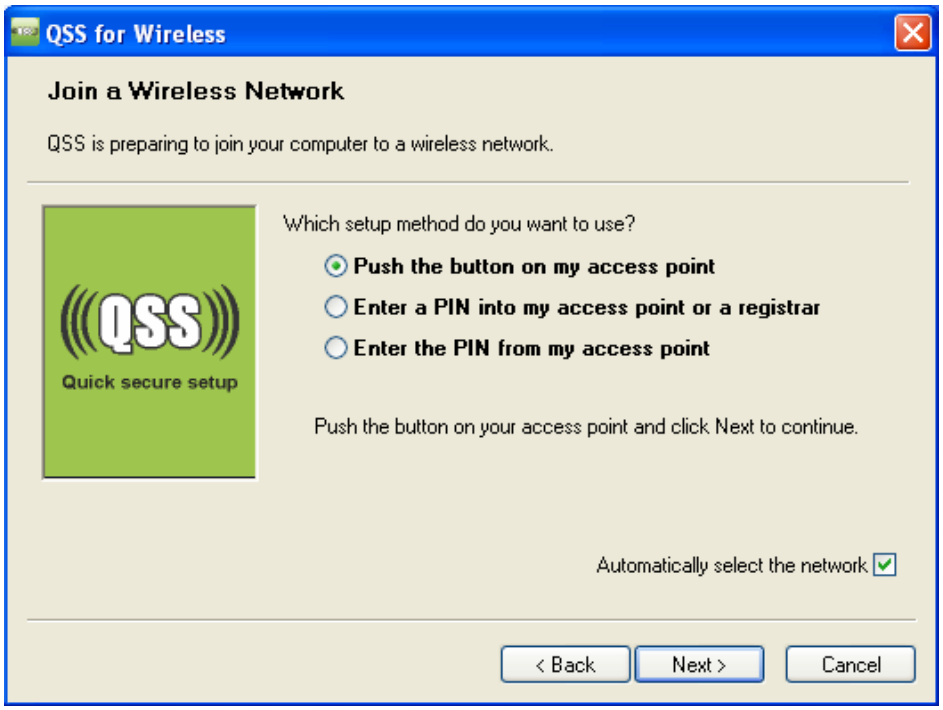


Figure 4-8

3. Then wait a minute until Figure 4-9 appears. Click **Finish** to complete the QSS configuration.

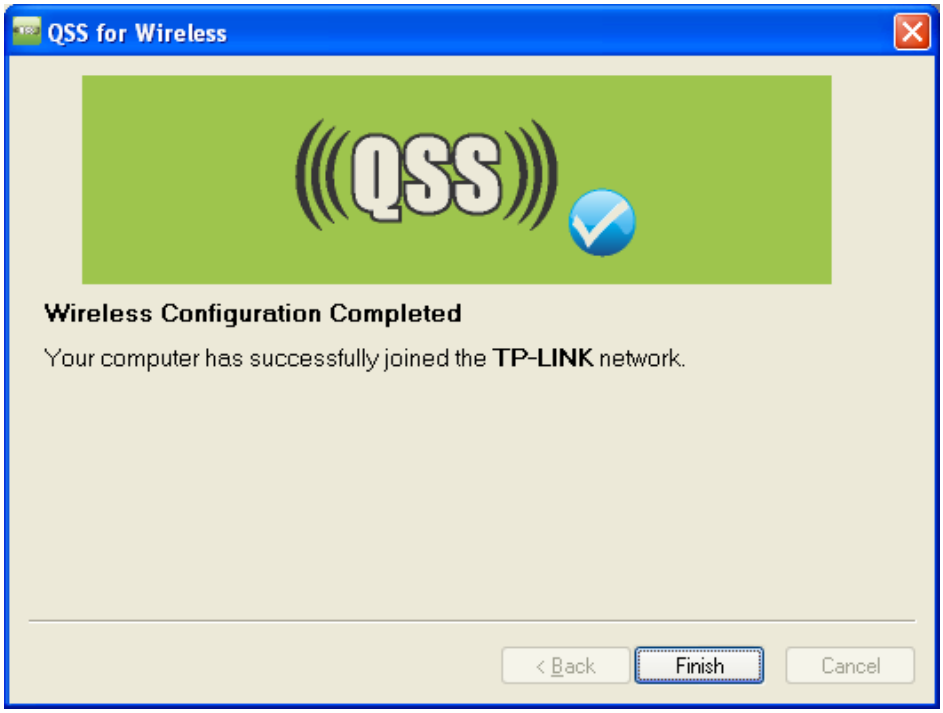


Figure 4-9

4.2 PIN method

There are two ways to configure the QSS by **PIN method**:

1) Enter a PIN into your AP device.

2) Enter the PIN from your AP device.

Following are the detailed configuration procedure of each way.

4.2.1 Enter a PIN into your AP device

1. Double click the icon  on the desktop to open the QSS Utility and then you can see the welcome screen shown as Figure 4-7 Click **Next** to continue. Figure 4-10 will appear. Select the second option, and you will see the PIN value of the adapter which is randomly generated. Click **Next**.

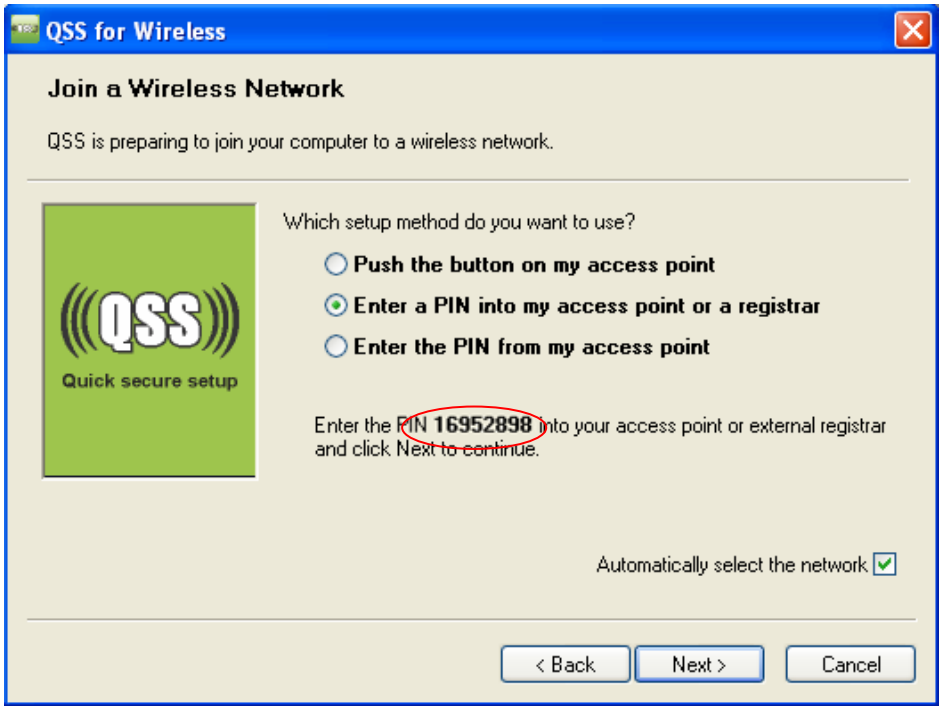


Figure 4-10

2. Open the Router's Web-based Utility and click **QSS** link on the left of the main menu. Then Figure 4-4 will appear. Click **Add device**, then you can see Figure 4-11 Select **Enter the new device's PIN** and enter the PIN value of the adapter shown in figure 4-9, click **Connect**.

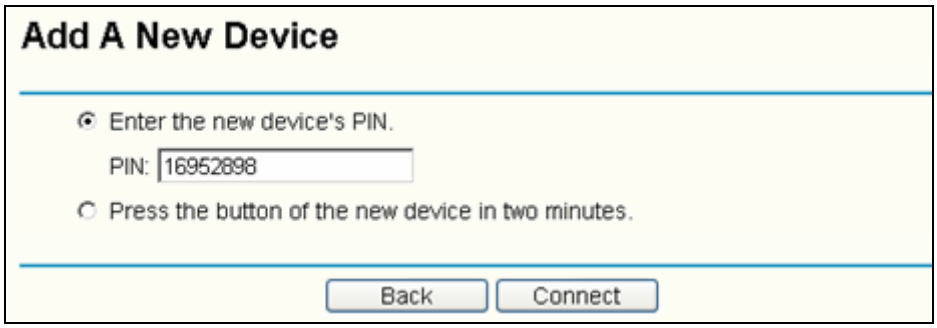


Figure 4-11

3. When Figure 4-9 appears, the QSS configuration is complete.

4.2.2 Enter the PIN from your AP device

1. Open the QSS Utility and you will see Figure 4-7. Click **Next** to continue. Then Figure 4-12 will appear. Select the third option and enter the PIN value which is labeled on the bottom of the Router. Click **Next**.

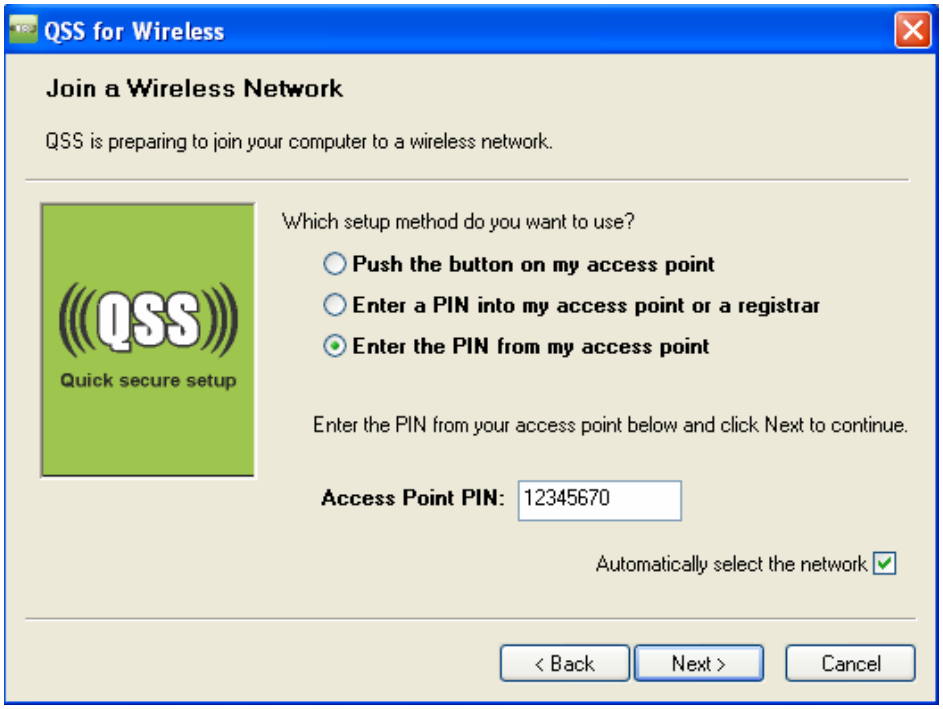


Figure 4-12

2. When Figure 4-9 appears, the QSS configuration is complete.

Note:

If you generate a new PIN code for your Router, please enter the new one instead.

Chapter 5. Example for Application

5.1 Configuration of PSP XLink Online game

Please ensure the software and hardware environments are well established before configuring. For hardware, at least a PC, a TL-WN821N Wireless N USB Adapter and a PSP device are needed. For software, the TL-WN821N Adapter driver should be properly installed.

Please operate as follows:

1. Connect the website of X-LINK http://www.teamxlink.co.uk to register, and download the latest software of X-LINK Kai.
2. Install the X-LINK Kai software, click **Start > Program > XLink Kai > Configure Kai**, and then set as Figure 5-1.

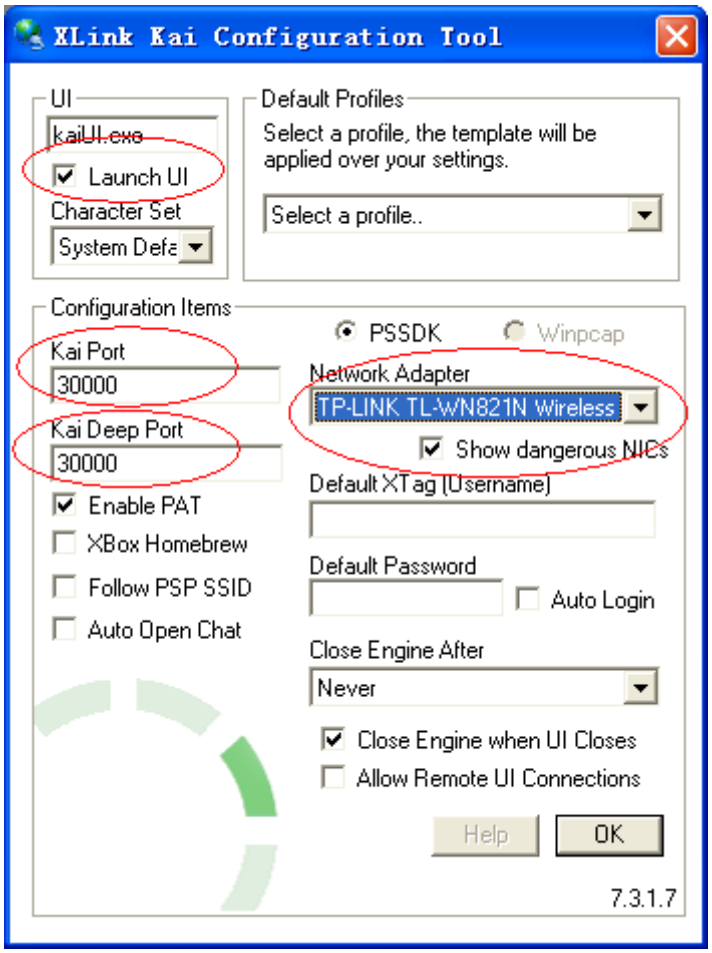


Figure 5-1

3. After completing the settings, please click **Start > Program > XLink Kai > Start Kai** to connect to XLink Kai.
4. Open the wireless mode of the PSP device, then start an internet game.
5. Click **Start > Control Panel > Network**.

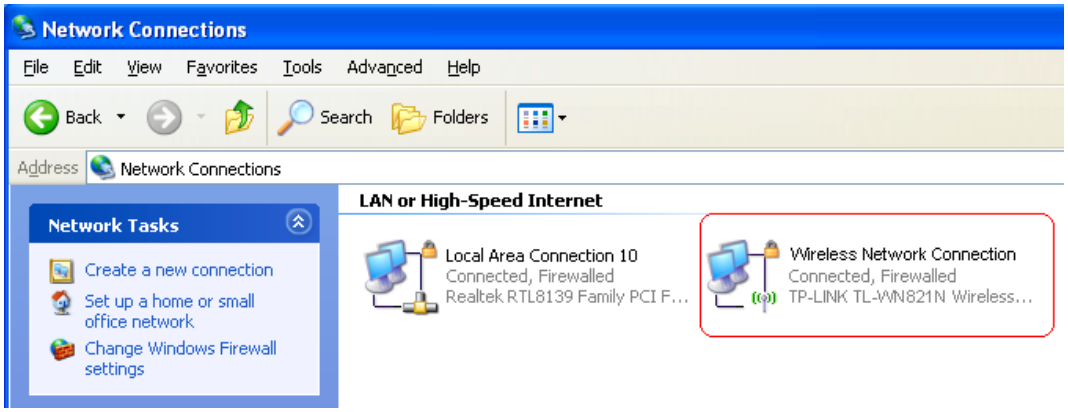


Figure 5-2

6. Right-click **Wireless Network Connection icon**  , and select **Properties.** In the following prompt page**,** highlight **Internet Protocol(TCP/IP)** and click **Properties**

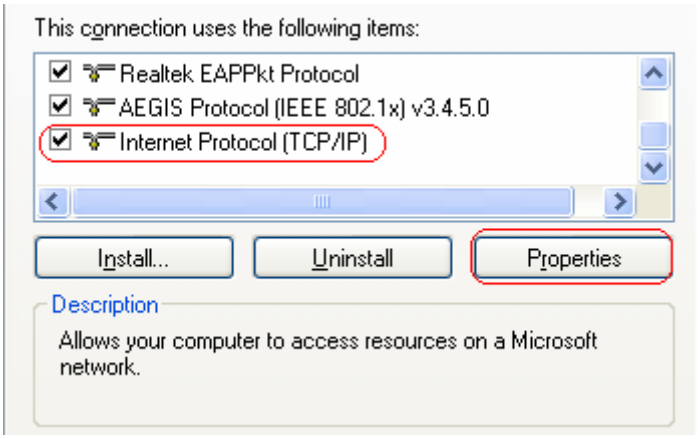


Figure 5-3

7. In the prompt page shown below, select **Use the Following IP Address**, and set the IP and Subnet mask. After completing setting, click **OK**.

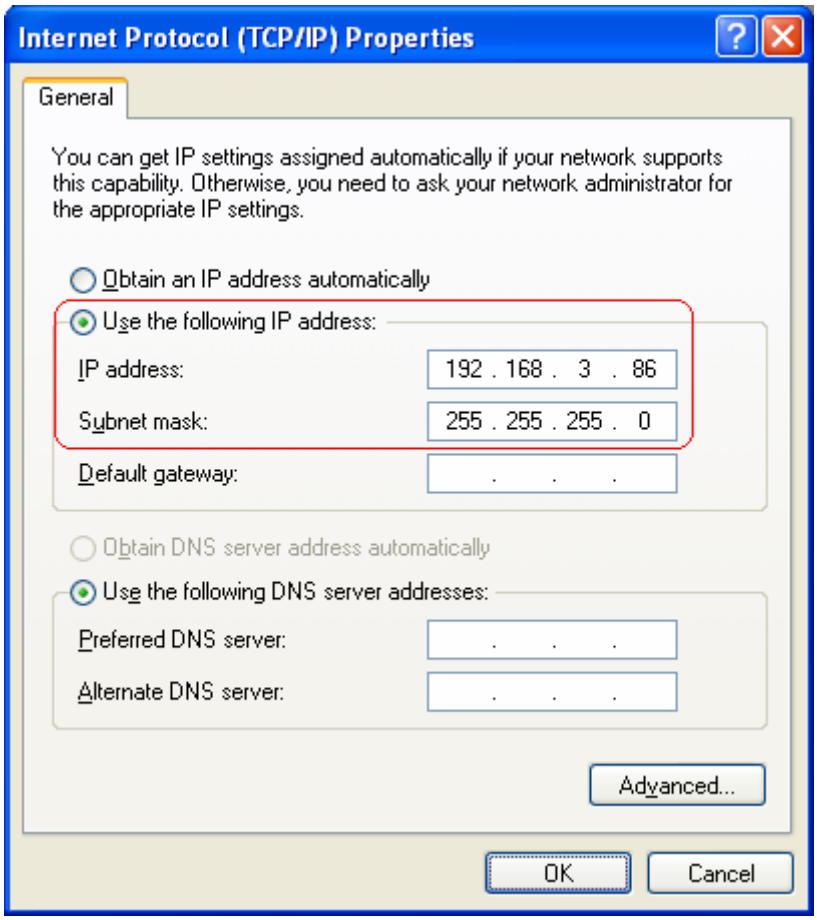


Figure 5-4

8. Launch TL-WN821N Wireless Utility, and click **Scan** in the page of “Profile Management”, then highlight the Network Name (SSID) beginning with “PSP” in the following “Available Infrastructure and Ad Hoc Networks” page, and click **Activate.**

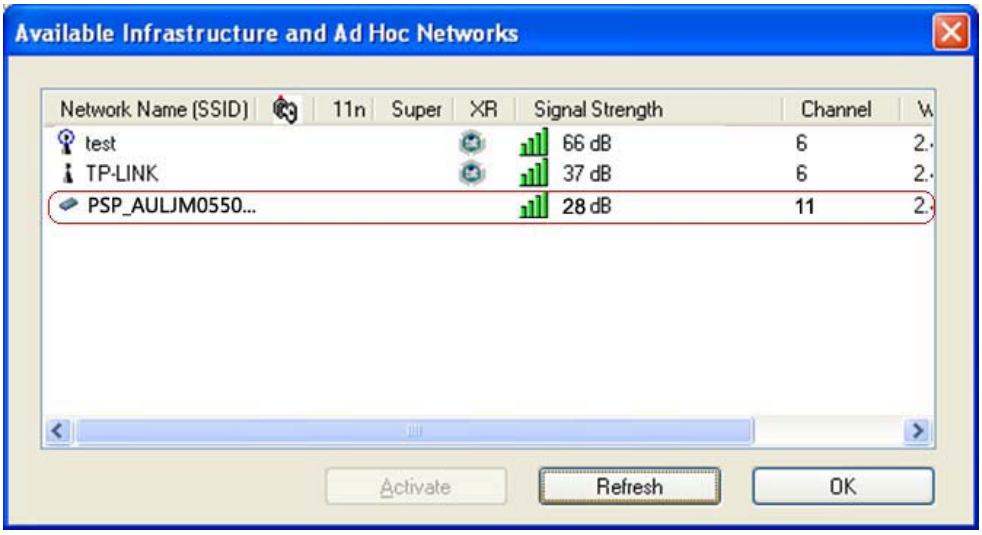


Figure 5-5

9. Check whether your PSP device is detected in the Diagnostics mode of Kai as Figure 5-6 shown: Click the icon first and then click the folder .

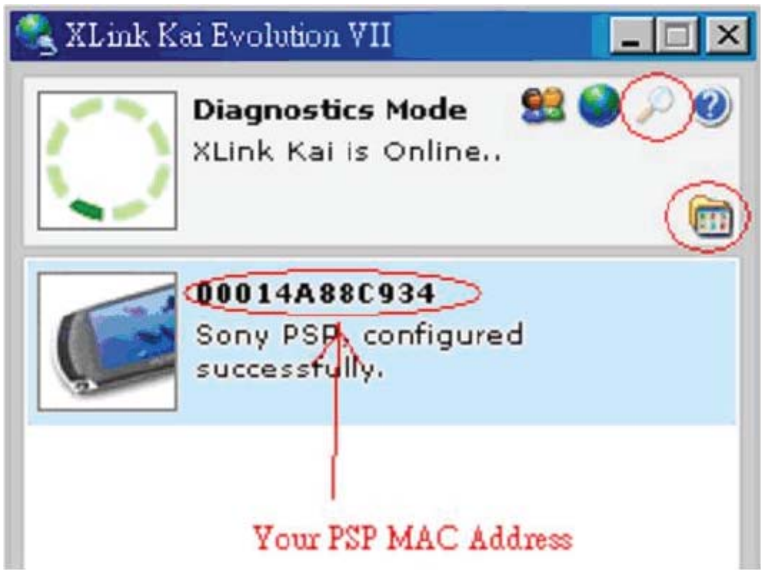


Figure 5-6

10. Click the icon on the right top corner to enter the Arena Mode, highlight the arena of your wanted game, and then join or start a new game.



Figure 5-7

Appendix A: Specifications

Normal	
Interface	USB 2.0 Interface
Standards	IEEE802.11b; IEEE802.11g; IEEE802.11n;
Operating System	Windows 2000, Windows XP, Windows Vista, Windows 7
Radio Data Rate	11b：1/2/5.5/11Mbps 11g：6/9/12/18/24/36/48/54Mbps 11n：Up to 300Mbps
Modulation	11b:CCK,QPSK,BPSK; 11g:OFDM; 11n: QPSK,BPSK,16-QAM,64-QAM
Media Access Protocol	CSMA/CA with ACK
Data Security	WPA/WPA2; 64/128/152-bit WEP; TKIP/AES
Frequency	2.4 ~ 2.4835GHz
Spread Spectrum	Direct Sequence Spread Spectrum (DSSS)
Safety & Emissions	FCC, CE

Environmental and Physical	
Operating Temp.	0℃~40℃ (32℉~104℉)
Storage Temp.	-20℃ ~ 70℃ (-4℉~158℉)
Humidity	10% ~ 95% RH, Non-condensing

Appendix B: Glossary

- **802.11b -** The 802.11b standard specifies a wireless product networking at 11 Mbps using direct-sequence spread-spectrum (DSSS) technology and operating in the unlicensed radio spectrum at 2.4GHz, and WEP encryption for security. 802.11b networks are also referred to as Wi-Fi networks.
- **802.11g -** specification for wireless networking at 54 Mbps using direct-sequence spread-spectrum (DSSS) technology, using OFDM modulation and operating in the unlicensed radio spectrum at 2.4GHz, and backward compatibility with IEEE 802.11b devices, and WEP encryption for security.
- **802.11n -** 802.11n builds upon previous 802.11 standards by adding MIMO (multiple-input multiple-output). MIMO uses multiple transmitter and receiver antennas to allow for increased data throughput via spatial multiplexing and increased range by exploiting the spatial diversity, perhaps through coding schemes like Alamouti coding. The Enhanced Wireless Consortium (EWC)[3] was formed to help accelerate the IEEE 802.11n development process and promote a technology specification for interoperability of next-generation wireless local area networking (WLAN) products.
- **Ad-hoc Network -** An ad-hoc network is a group of computers, each with a Wireless Adapter, connected as an independent 802.11 wireless LAN. Ad-hoc wireless computers operate on a peer-to-peer basis, communicating directly with each other without the use of an access point. Ad-hoc mode is also referred to as an Independent Basic Service Set (IBSS) or as peer-to-peer mode, and is useful at a departmental scale or SOHO operation.
- **DSSS -** (**D**irect-**S**equence **S**pread **S**pectrum) **-** DSSS generates a redundant bit pattern for all data transmitted. This bit pattern is called a chip (or chipping code). Even if one or more bits in the chip are damaged during transmission, statistical techniques embedded in the receiver can recover the original data without the need of retransmission. To an unintended receiver, DSSS appears as low power wideband noise and is rejected (ignored) by most narrowband receivers. However, to an intended receiver (i.e. another wireless LAN endpoint), the DSSS signal is recognized as the only valid signal, and interference is inherently rejected (ignored).
- **FHSS -** (**F**requency **H**opping **S**pread **S**pectrum) **-** FHSS continuously changes (hops) the carrier frequency of a conventional carrier several times per second according to a pseudo-random set of channels. Because a fixed frequency is not used, and only the transmitter and receiver know the hop patterns, interception of FHSS is extremely difficult.
- **Infrastructure Network -** An infrastructure network is a group of computers or other devices, each with a Wireless Adapter, connected as an 802.11 wireless LAN. In infrastructure mode, the wireless devices communicate with each other and to a wired network by first going through an access point. An infrastructure wireless network connected to a wired network is referred to as a Basic Service Set (BSS). A set of two or more BSS in a single network is referred to as an Extended Service Set (ESS). Infrastructure mode is useful at a corporation scale, or when it is necessary to connect the wired and wireless networks.
- **Spread Spectrum -** Spread Spectrum technology is a wideband radio frequency technique

developed by the military for use in reliable, secure, mission-critical communications systems. It is designed to trade off bandwidth efficiency for reliability, integrity, and security. In other words, more bandwidth is consumed than in the case of narrowband transmission, but the trade off produces a signal that is, in effect, louder and thus easier to detect, provided that the receiver knows the parameters of the spread-spectrum signal being broadcast. If a receiver is not tuned to the right frequency, a spread-spectrum signal looks like background noise. There are two main alternatives, Direct Sequence Spread Spectrum (DSSS) and Frequency Hopping Spread Spectrum (FHSS).

- **SSID -** A **S**ervice **S**et **Id**entification is a thirty-two character (maximum) alphanumeric key identifying a wireless local area network. For the wireless devices in a network to communicate with each other, all devices must be configured with the same SSID. This is typically the configuration parameter for a wireless PC card. It corresponds to the ESSID in the wireless Access Point and to the wireless network name. *See also* Wireless Network Name and ESSID.

- **WEP -** (**W**ired **E**quivalent **P**rivacy) **-** A data privacy mechanism based on a 64-bit or 128-bit or 152-bit shared key algorithm, as described in the IEEE 802.11 standard. To gain access to a WEP network, you must know the key. The key is a string of characters that you create. When using WEP, you must determine the level of encryption. The type of encryption determines the key length. 128-bit encryption requires a longer key than 64-bit encryption. Keys are defined by entering in a string in HEX (hexadecimal - using characters 0-9, A-F) or ASCII (American Standard Code for Information Interchange – alphanumeric characters) format. ASCII format is provided so you can enter a string that is easier to remember. The ASCII string is converted to HEX for use over the network. Four keys can be defined so that you can change keys easily.

- **Wi-Fi -** A trade name for the 802.11b wireless networking standard, given by the Wireless Ethernet Compatibility Alliance (WECA, see http://www.wi-fi.net), an industry standards group promoting interoperability among 802.11b devices.

- **WLAN -** (**W**ireless **L**ocal **A**rea **N**etwork) - A group of computers and associated devices communicate with each other wirelessly, which network serving users are limited in a local area.

- **WPA -** (**W**i-Fi **P**rotected **A**ccess) **-** A wireless security protocol use TKIP (Temporal Key Integrity Protocol) encryption, which can be used in conjunction with a RADIUS server.